



MATCHING NATURE

Linking tooth properties
with material selection

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BEHAVIORAL OBJECTIVES

1. **Identify** the key properties of natural teeth, including mechanical, physical, chemical, and biological, and explain their role in tooth function and longevity.
2. **Differentiate** between enamel and dentin by describing their distinct structural, mechanical, chemical, and biological properties, and explain how these differences influence overall tooth behavior.
3. **Explain** how natural tooth properties respond to challenges in the oral environment (such as acid attack, mechanical loading, and thermal cycling) and how these responses influence susceptibility to disease and the requirements for restorative material selection.
4. **Compare and contrast** restorative materials with natural teeth across these property domains, evaluating where they align, where they diverge, and how these matches or mismatches impact clinical outcomes such as fracture resistance, wear, esthetics, and biocompatibility.
5. **Apply** an evidence-based decision framework to **select** restorative materials and techniques that best approximate the multidimensional properties of teeth, integrating biological, mechanical, chemical, and patient-specific considerations.

BEHAVIORAL OBJECTIVES

1. Identify the key properties of natural teeth, including mechanical, physical, chemical, and biological, and explain their role in tooth function and longevity.

Behavioral Objective 1

Search Strategy

P – Natural human teeth

I – Mechanical, physical, chemical, and biologic properties

C – None

O – Tooth function and longevity

Databases

PubMed

Google Scholar

NCBI

ChatGPT

Keywords: mechanical properties, physical properties, chemical properties, biologic properties, natural teeth, natural tooth function, tooth function

Results:

- 23,700 unfiltered
- 18,600 filtered for date range 2014–2026
- 8,800 filtered for reviews

Highest level of evidence: review

Natural Tooth Composition

- Natural teeth composed of **enamel**, **dentin**, **cementum**, and **pulp**
- **Enamel:** outermost layer, highly mineralized with hydroxyapatite crystals
 - 96% inorganic (HA), 1% organic, 3% water
- **Dentin:** beneath enamel, the bulk of the tooth, less mineralized, surrounds pulp
 - 70% inorganic, 20% organic (collagen), 10% water



Important Definitions

Elastic modulus: the ratio of stress: strain under an elastic state

Hardness: ability of a solid material to resist deformation

Fracture toughness: how well the tooth can resist growth of cracks under stress

Isotropic: same properties in different directions

Anisotropic: different properties in different directions

Stress: force per unit area

Strain: amount of deformation when stress is applied

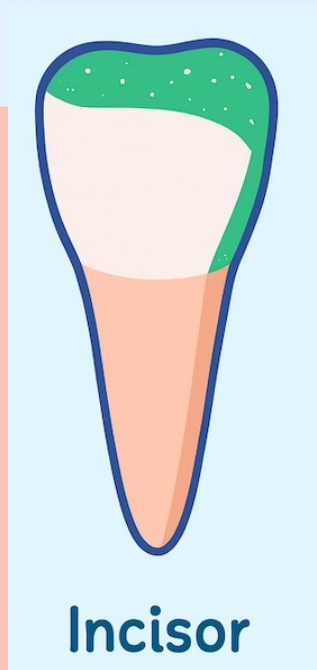
Mechanical Properties

- Teeth require these mechanical properties in order to function over an individual's lifetime
- **Hardness** – to incise, lacerate, and masticate food
- **Elastic modulus** – to maintain natural shape and occlusion during mastication forces
- **Fracture toughness** – to prevent cracks during biting and grinding forces

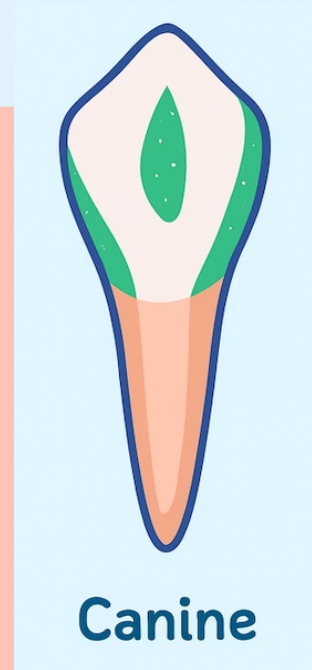


Mechanical Properties

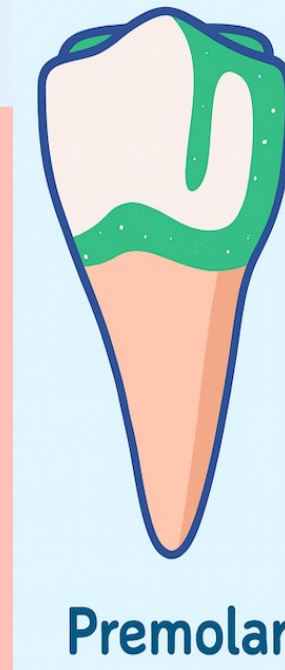
- Morphologic properties depending on the type of tooth



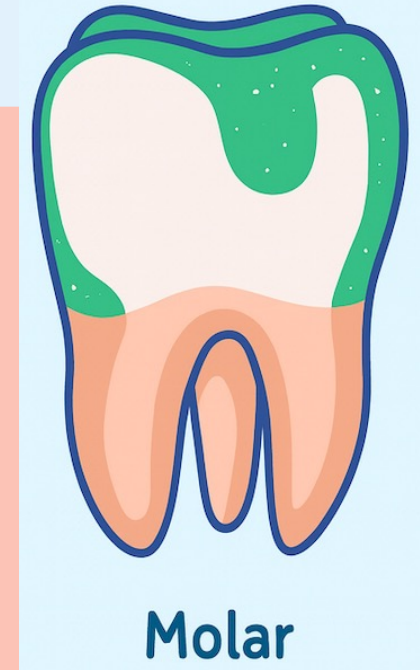
- Sharp incisal edges
- Designed for cutting and incising



- Pointed incisal edge
- Designed for tearing



- Broad, blunt cusps
- Designed for grinding and crushing



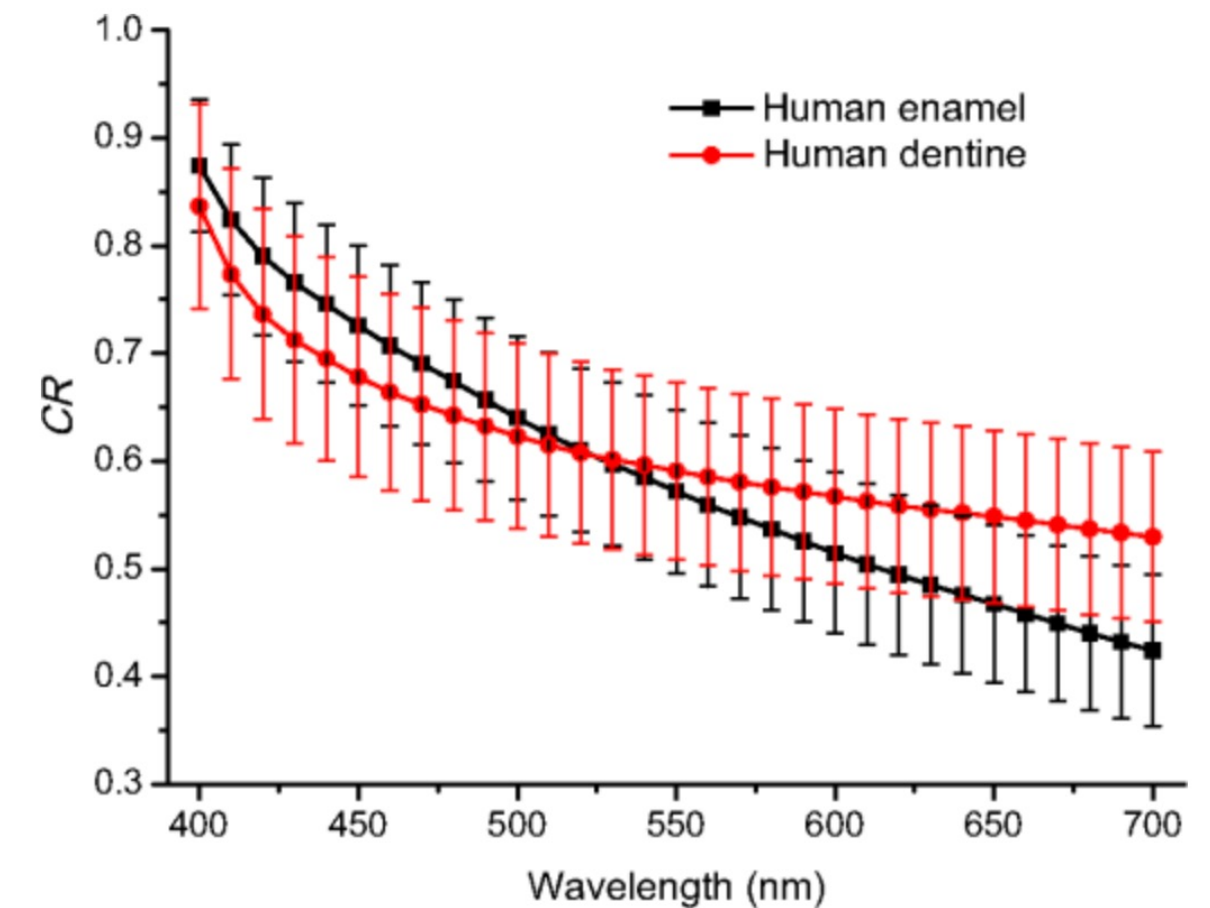
Physical Properties

- **Translucency and opacity**
 - The different shades of natural teeth result from enamel's transparency and dentin's opacity
 - Organic components of natural teeth absorb more light
 - Dentin has higher organic component than enamel
 - **Higher CR = more opaque**



Fig. 2 [Download](#)

Wavelength-dependent contrast ratio (CR) of human enamel and human dentine.³³



Physical Properties

- **Thermal conductivity and thermal expansion**
- **Thermal conductivity:** ability of a material to transfer heat
- **Thermal expansion:** how much a material increases in volume when heated

TABLE 4.10 Thermal Conductivity (K) of Various Materials

Material	Thermal Conductivity	
	cal/sec/cm ² /° C/cm	J/sec/cm ² /° C/cm
METALS		
Silver	1.006	4.21
Copper	0.918	3.84
Gold	0.710	2.97
Platinum	0.167	0.698
Dental amalgam	0.055	0.23
Mercury	0.020	0.084
NONMETALS		
Gypsum	0.0031	0.013
Zinc phosphate cement	0.0028	0.012
Resin composite	0.0026	0.011
Porcelain	0.0025	0.010
Enamel	0.0022	0.0092
Dentin	0.0015	0.0063
Zinc oxide–eugenol cement	0.0011	0.0046
Acrylic resin	0.0005	0.0021
Beeswax	0.00009	0.0004

TABLE 4.13 Linear Coefficient (α) of Thermal Expansion of Various Materials

Material	Coefficient ($\times 10^{-6}/^{\circ}\text{C}$)
Inlay waxes	350–450
Silicone impression material	210
Polysulfide impression material	140
Pit and fissure sealants	71–94
Acrylic resin	76.0
Mercury	60.6
Resin composites	14–50
Zinc oxide–eugenol cement	35
Amalgam	22–28
Silver	19.2
Copper	16.8
Gold	14.4
Porcelain	12.0
Tooth (crown portion)	11.4
Glass ionomer (type 2)	10.2–11.4

Chemical Properties

- **Organic** polymer framework: carbon, nitrogen, oxygen, hydrogen
- **Inorganic** precipitates: calcium, phosphate, and trace elements
 - **Calcium:** main component of teeth and bone, most abundant ion in enamel, protective against demineralization
 - **Phosphate:** also involved in mineralization, determines hardness and mechanical resistance

NO significant correlation of differing chemical composition between teeth types
(Sarna-Boś et. al)

Chemical Properties

- Solubility of natural teeth
- Trace elements can enter through food, water, or environment exposure
- **Carbonate** substitution --> increases solubility
- **Fluoride** substitution --> decreases solubility --> lowers solubility by at least 3x magnitude --> withstands a lower pH environment



Biological Properties

- Bioadhesion in the Oral Cavity
 - Teeth covered by the **acquired pellicle**: an electron-dense layer containing adsorbed salivary proteins and macromolecules from GCF, blood, bacteria, mucosa, and the individual's diet



Acquired pellicle contains antibacterial proteins (lysozymes and peroxidases) and lubricants (mucin) to protect the teeth.

Biological Properties

- Teeth crucial for
 - Mechanically breaking down food
 - Speech
 - Facial esthetics
 - Jawbone structure



BEHAVIORAL OBJECTIVES

2. Differentiate between enamel and dentin by describing their distinct structural, mechanical, chemical, and biological properties, and explain how these differences influence overall tooth behavior.

Behavioral Objective 2

Search Strategy

P – Natural human teeth

I – Mechanical, physical, chemical, and biologic properties

C – None

O – Tooth function and longevity

Databases

PubMed

Google Scholar

NCBI

ChatGPT

Keywords: mechanical properties, physical properties, chemical properties, biologic properties, natural teeth, natural tooth function, tooth function

Results:

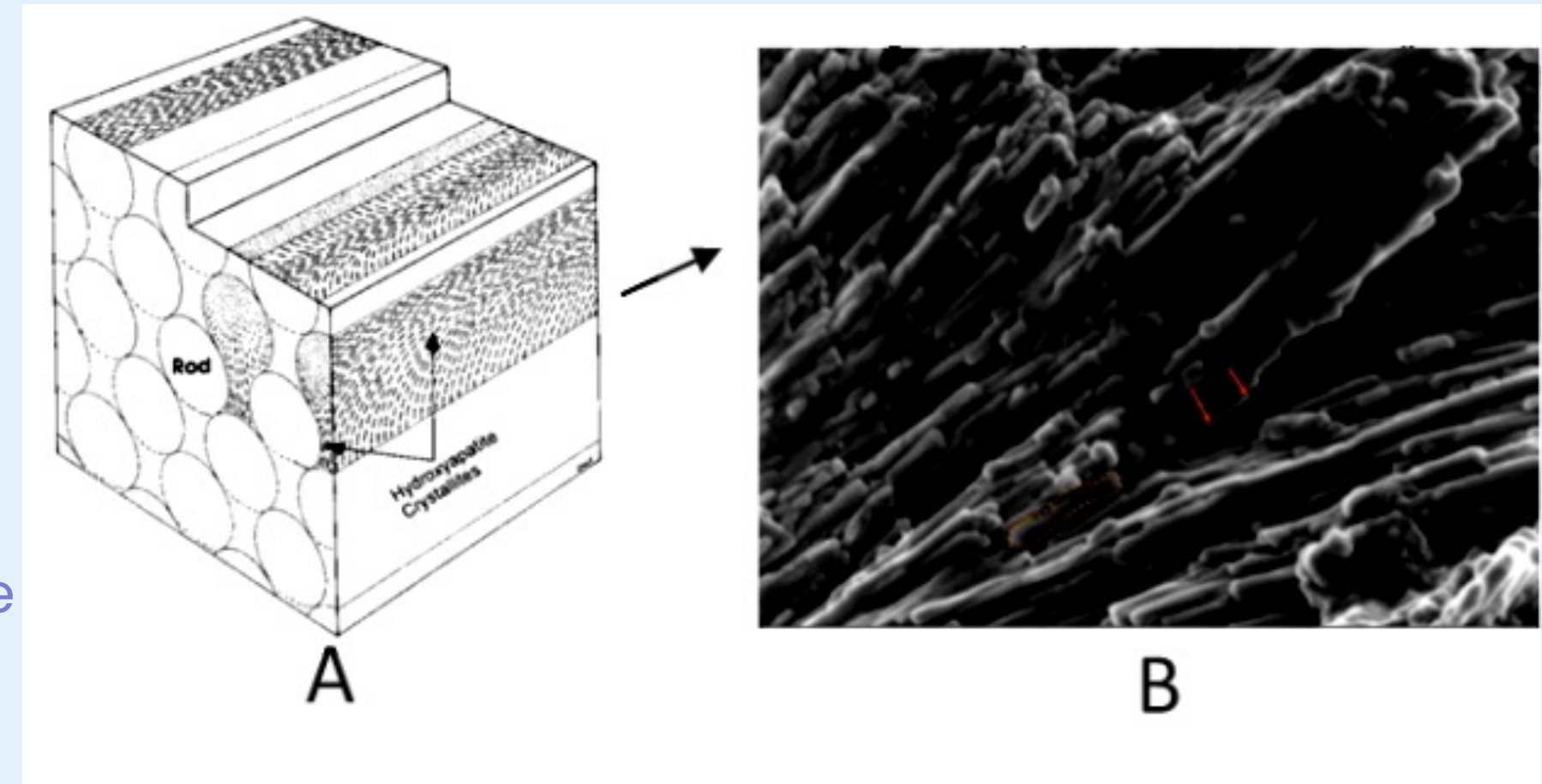
- 23,700 unfiltered
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Highest level of evidence: review

Structural Differences

Enamel

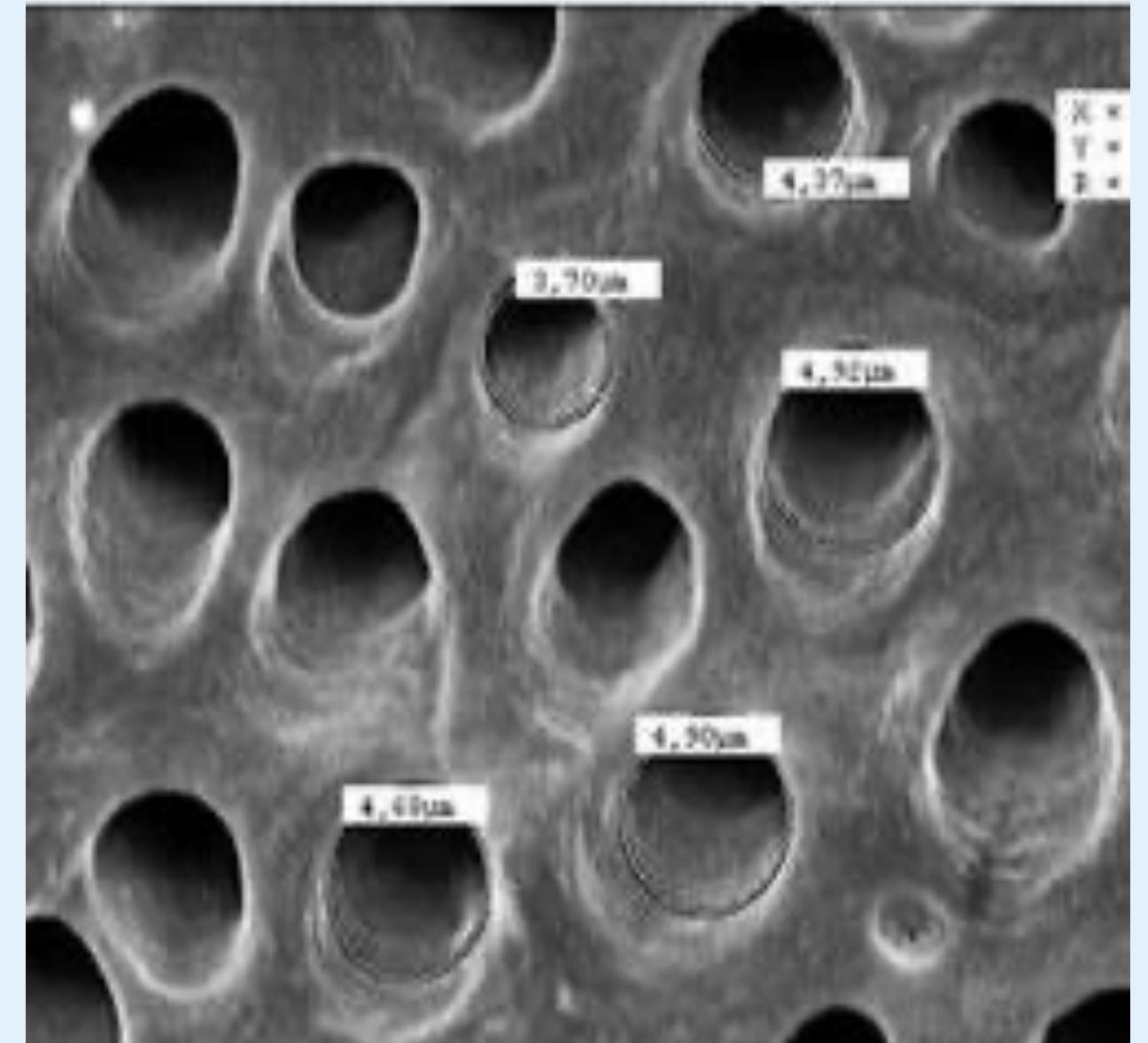
- Composed of enamel rods and prisms tightly packed
- Rods: key-hole like structures consisting of hexagonal hydroxyapatite crystals
- Different angulation of the crystals leads to anisotropy of the material
- Hunter-Schreger bands: interwoven inner enamel. Made of dozen enamel rods (largest structural level)
- 96% hydroxyapatite
- Outer enamel has higher hardness and higher elastic modulus, higher calcium concentration



Structural Differences

Dentin

- Contains hydrated dentinal tubes
- Pulp → DEJ/CEJ
- Tubules contain
 - Odontoblast processes and dentinal fluid
- Surrounded by peritubular (Highly mineralized ring around lumen) and intertubular dentin (collagen-rich mineralized matrix)
- Middle dentin has higher hardness and elastic modulus than outer and inner portion



Zhang, YR., Du, W., Zhou, XD. *et al.* (2014)

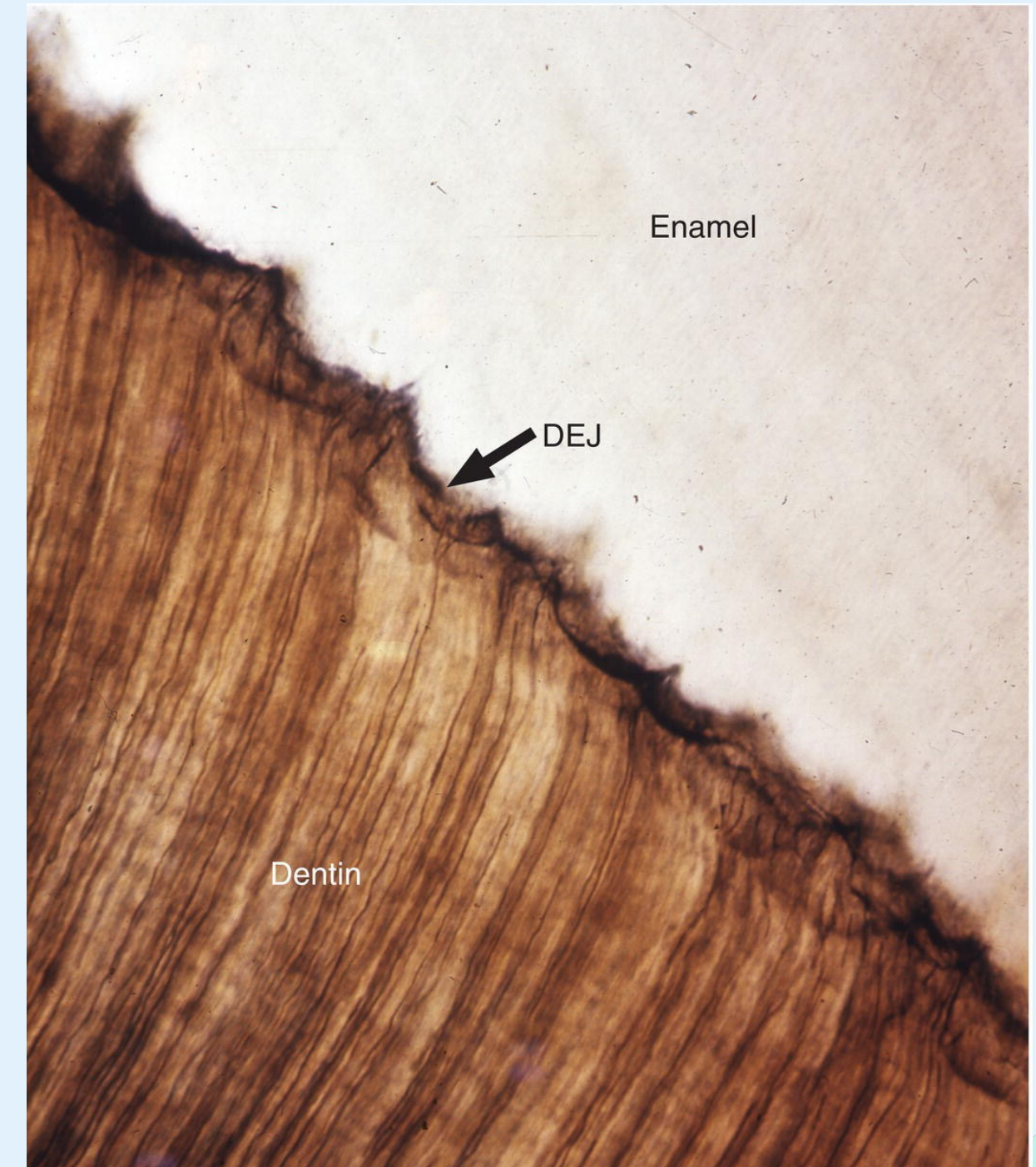
H. Zhao, Y. Li, X. Zhang, J. Xu, L. Chen, & Z. Wu. (2022).

Gemini

Structural Differences

Dental enamel junction

- Scalloped shape
- Convexities toward dentin and concavities toward enamel
- Increased surface area, increases bond between both layers and allows crack deflection
- Width of DEJ highly contentious



Xu, C., Yao, X., Walker, M. P., & Wang, Y. (2009).

Marshall, S. J., Balooch, M., Habelitz, S., Balooch, G., Gallagher, R. R., & Marshall, G. W. Jr. (2003).

White, S. N., Miklus, V. G., Chang, P. P., Caputo, A. A., Fong, H., Sarikaya, M., Luo, W., Paine, M. L., & Snead, M. L. (2005).

Hand, A. R. (n.d.). *Enamel*. In *Fundamentals of Oral Histology and Physiology* (Chapter 4). Gemini

Mechanical Differences

Enamel: The Hard Shell

Extremely hard and stiff . Optimized for **wear resistance** .

Hardness (Vickers): **274.8** **4.2x Higher**

Max Compressive Stress: **⚠ 62.2 MPa**

Max Compressive Strain: **💧 4.5 %**

Elastic Modulus (Stiffness): **1338.2 MPa**

Fracture Toughness: **Low (0.67–3.93 MPa√m)**

Dentin: The Shock Absorber

Tougher and more elastic . Its collagen-tubule structure provides crucial support and **flexibility** .

Hardness (Vickers): **65.6**

Max Compressive Stress: **👉 193.7 MPa**

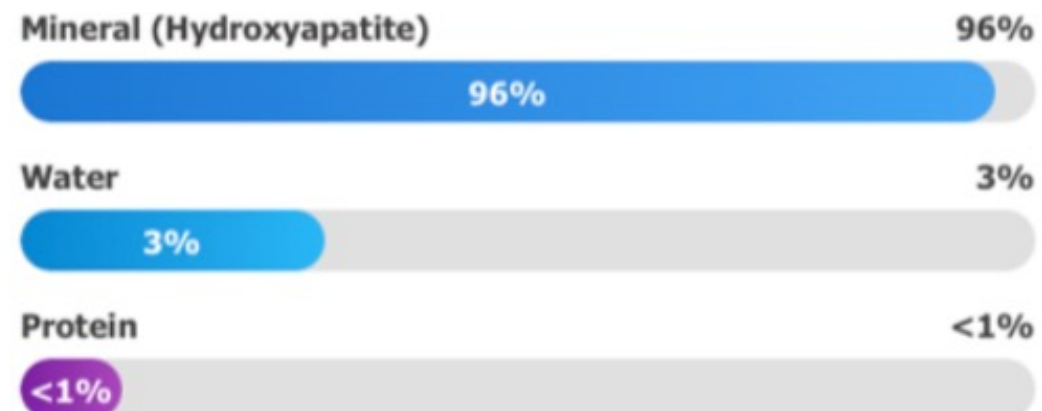
Max Compressive Strain: **11.9 %**

Elastic Modulus (Stiffness): **1653.7 MPa**

Fracture Toughness: **High (2–3 MPa√m)**

Chemical Differences

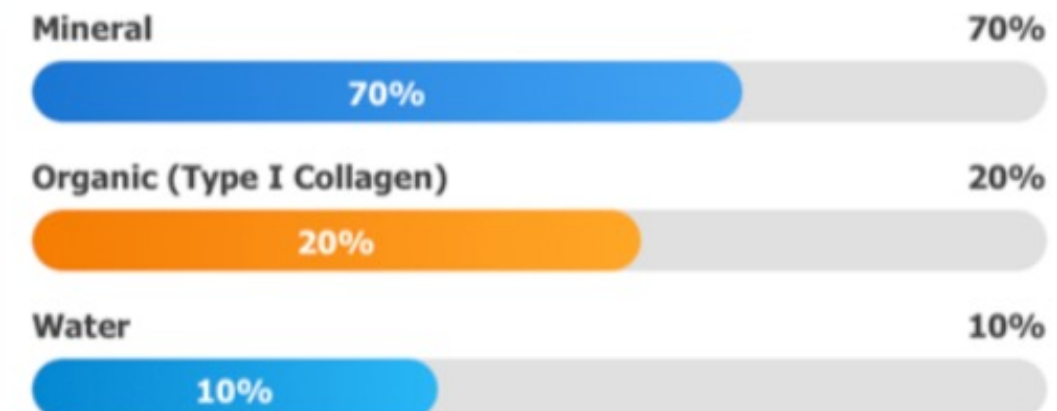
Enamel



Structural Gradient: Outer enamel has higher concentration of hydroxyapatite than inner enamel

⚠ Critical pH: 5.5

Dentin



Environmental Impact: Environment matters

Hydration Effect: Hydrated dentin is anisotropic and has lower hardness and elastic modulus than dehydrated dentin

Solubility: More soluble due to carbonate content

Age Factor: Erosion increases with age

⚠ Critical pH: 6.2

Fejerskov, O., Nyvad, B. & Kidd. (John Wiley & Sons, 2015).

Chun, K., Choi, H., & Lee, J. (2014)

Shellis and Lussi (2014)

Biological Differences

Enamel

Cellular Characteristics

 **Acellular:** Contains no cells after formation

 **Avascular:** No blood supply

 **Aneural:** No nerve innervation

Composition

 **<1% organic composition**

 **Low Permeability**

Dentin

Cellular Structure

 **Cellular Components:** Composed of **odontoblastic processes** extending through the dental tubules

 **Vascularity:** **Indirect vascularity** due to proximity to the pulp

Metabolic Activity

 **Metabolic Process:** Low metabolic activity via **odontoblastic secretion** of phosphoproteins and collagens

Composition

 **Organic Content:** **20% Type I collagen** and non-collagenous proteins

 **High Permeability**

Enamel Fracture Resistance

Enamel

Fracture Pattern

Fracture Path: Cracks propagate along **rod sheath** boundaries

Protective Gradient

Structural Defense: **Inner enamel's higher toughness** increases crack resistance as the crack propagates toward the pulp

This gradient structure provides progressive protection, with the inner layers becoming increasingly resistant to crack advancement

Dentinal Fracture Resistance



Dentin

⚡ Structural Advantage

Anisotropic Nature: Guides cracks to propagate in **different directions and angles**, effectively distributing force throughout the structure

🔍 Fracture Pattern

Coronal Dentin: Fractures along **peritubular dentin**



Dentin Toughening Mechanisms

External Toughening

Broken pieces of dentin form bridges after crack formation, preventing further crack propagation

Internal Toughening

Microcracking: Tiny cracks appear near the main crack tip, absorbing energy

Plastic Deformation: Dentin undergoes bending and stretching, dissipating energy before catastrophic failure

Influences on Tooth Behavior

Enamel

- High hydroxyapatite concentration increases enamel wear resistance. Used as a grinding surface
- Anisotropy of rods aids in toughness.
- Rod decussation increases wear resistance
- Permanent damage

Dentin

- Collagen composition gives flexibility and toughness. Used as a shock absorber
- Anisotropy aids in toughness as well
- Reparative and tertiary dentin.

Zhang, YR., Du, W., Zhou, XD. et al. (2014)

Cohen, S. R., Apter, N., Jesse, S., Kalinin, S., Barlam, D., Peretz, A. I., Ziskind, D., & Wagner, H. D. (2008).

Influences on Tooth Behavior

Coefficient of Thermal Expansion:

- Enamel = $16.96 \times 10^{-6}/^{\circ}\text{C}$
- Dentin = $11.90 \times 10^{-6}/^{\circ}\text{C}$
- Consequence
 - Differential expansion Hot/Cold foods
 - Formation of microcracks

Translucency:

- Total Light Transmission (550nm)
 - Enamel = 60–80%
 - Dentin = 20–40%
- Translucency parameter
 - Enamel TP = 15–22
 - Dentin TP = 4–10
- Scattering (632nm)
 - Enamel = $10\text{--}30\text{cm}^{-1}$
 - Dentin = $150\text{--}300\text{cm}^{-1}$
- Absorption
 - Enamel = $0.01\text{--}0.1\text{cm}^{-1}$
 - Dentin = $0.5\text{--}2\text{cm}^{-1}$

Brodelt, R. H. W., O'Brien, W. J., Fan, P. L., Frazier-Dib, J. G., & Yu, R. (1981).

Ryan, E. A., Tam, L. E., & Linden, D. (2010).

ten Bosch, J. J., & Coops, J. C. (1995).

Xu, H. C., Liu, W. Y., & Wang, T. (1989).

BEHAVIORAL OBJECTIVES

3. Explain how natural tooth properties respond to challenges in the oral environment (such as acid attack, mechanical loading, aging, and thermal cycling) and how these responses influence susceptibility to disease and the requirements for restorative material selection.

Behavioral Objective 3

Search Strategy

P – Natural human teeth

I – Challenges in oral environment (acid attack, mechanical loading, aging, thermal cycling)

C – None

O – Susceptibility to disease

Databases

PubMed

ChatGPT

Keywords: enamel, dentin, environment, mechanical properties, aging, carbonated drinks

- (enamel OR dentin) AND environment AND properties
 - 340 results
- Enamel AND carbonated drink
 - 40 results
- (enamel OR dentin) AND (age or aging) AND mechanical

Highest level of evidence: Systematic review

Oral Environment Challenges

- **Acid attack**
 - Carbonated drinks
 - Sugar
- **Mechanical loading**
 - Bruxism, chewing
 - Fracture and fatigue
- **Aging**
 - Changing tooth properties
- **Thermal Cycling**
 - Variety of temperatures
- **Xerostomia**
 - Reduced salivary flow



Acid Attack

- **Dental-erosion**
 - Chemical-mechanical process
 - Reduction of hard enamel by removing the softened layer attacked by acids
 - Increases surface roughness
 - After 60 mins of exposure
 - Patients aware of dental erosion consumed fewer acidic drinks
- **Critical pH**
 - **Enamel:** 5.5
 - **Dentin:** 6.8



(A)



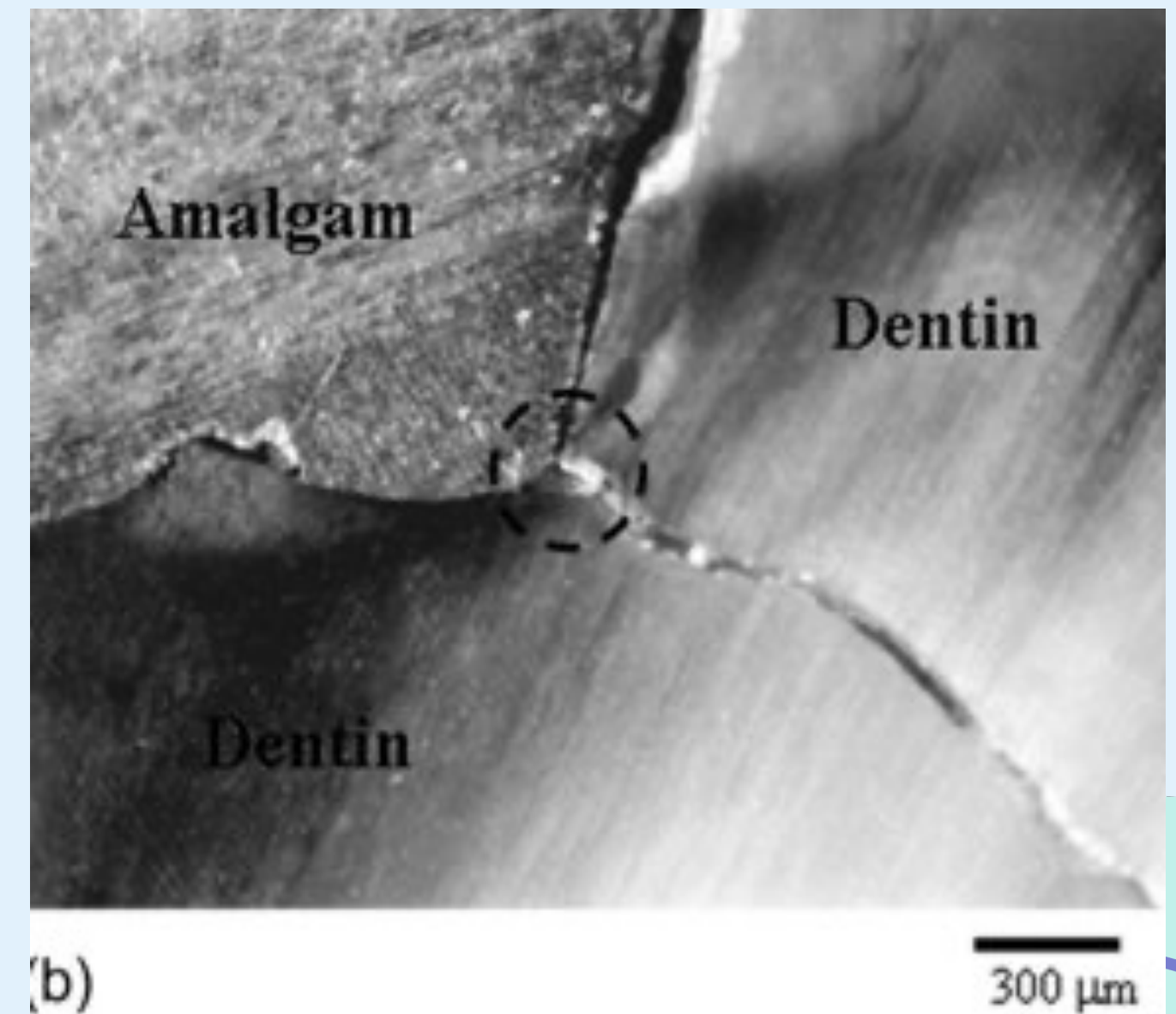
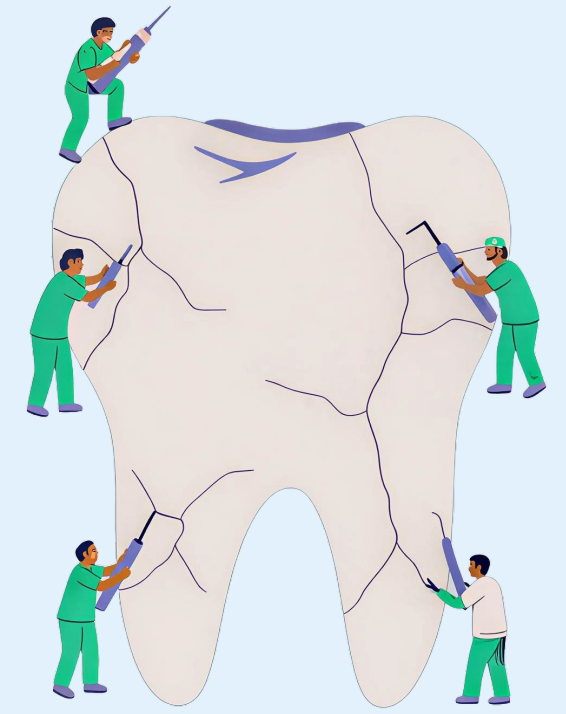
(B)



(C)

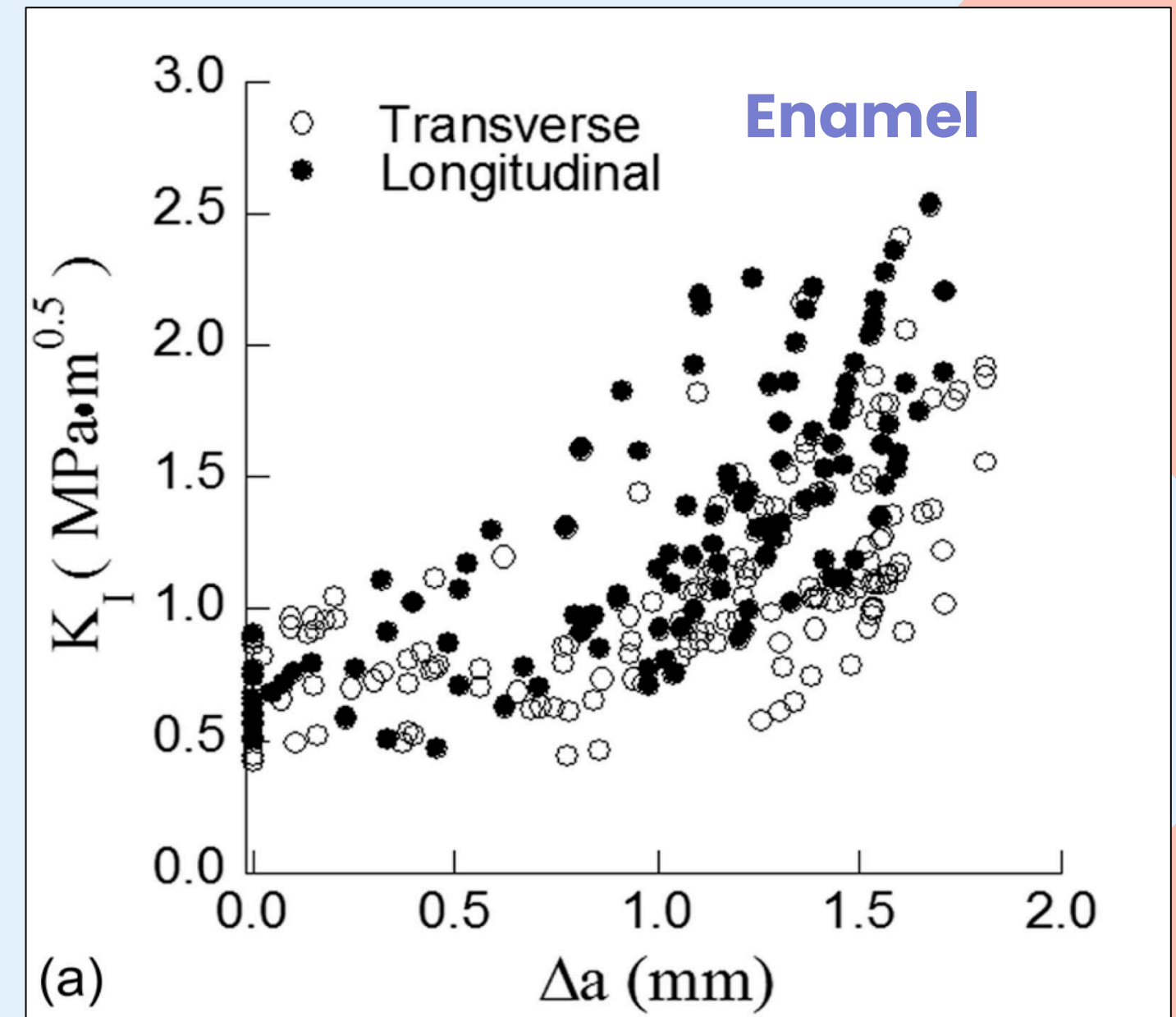
Mechanical Loading

- Cracks occur at interface between restoration and tooth
 - Often crack can't be seen
- Chipping and cracking of restorations and teeth lead to marginal failures
 - More **susceptible to secondary caries**
- Repeated **microscale** contacts can wear the tooth surface
- **Macroscale** contacts (100–1000N) can cause cracks



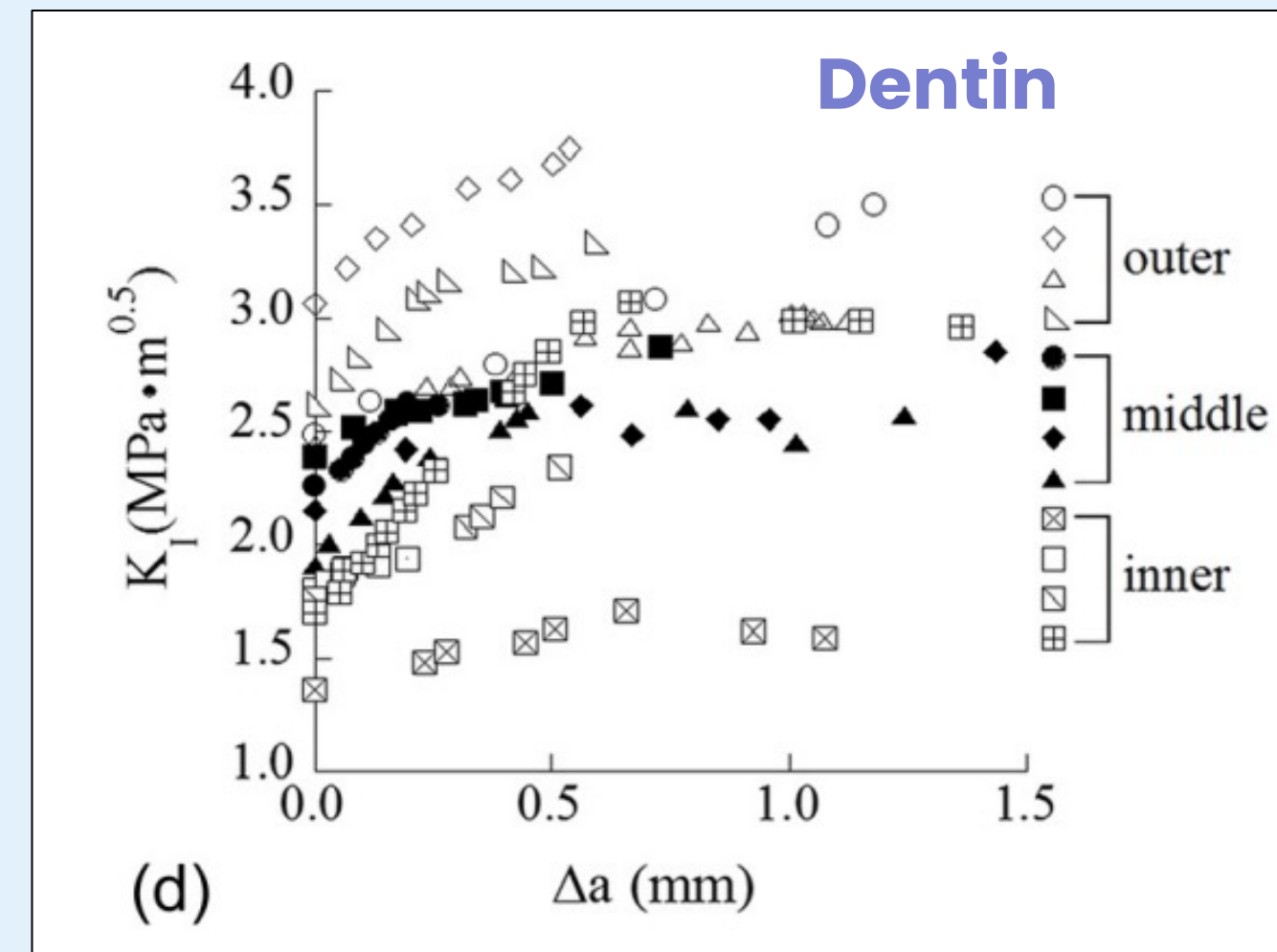
Mechanical Loading

- **Fracture resistance of teeth**
 - Most cracks limited to enamel and don't affect tooth function
 - Studies have wide variance of fracture resistance
 - **Inherent flaws** contribute to tooth cracks and failures
 - Crack growth resistance
 - Enamel growth toughness is less in the transverse direction
 - Causes cracks to work their way around the tooth instead of towards the DEJ (DEJ is next line of defense)



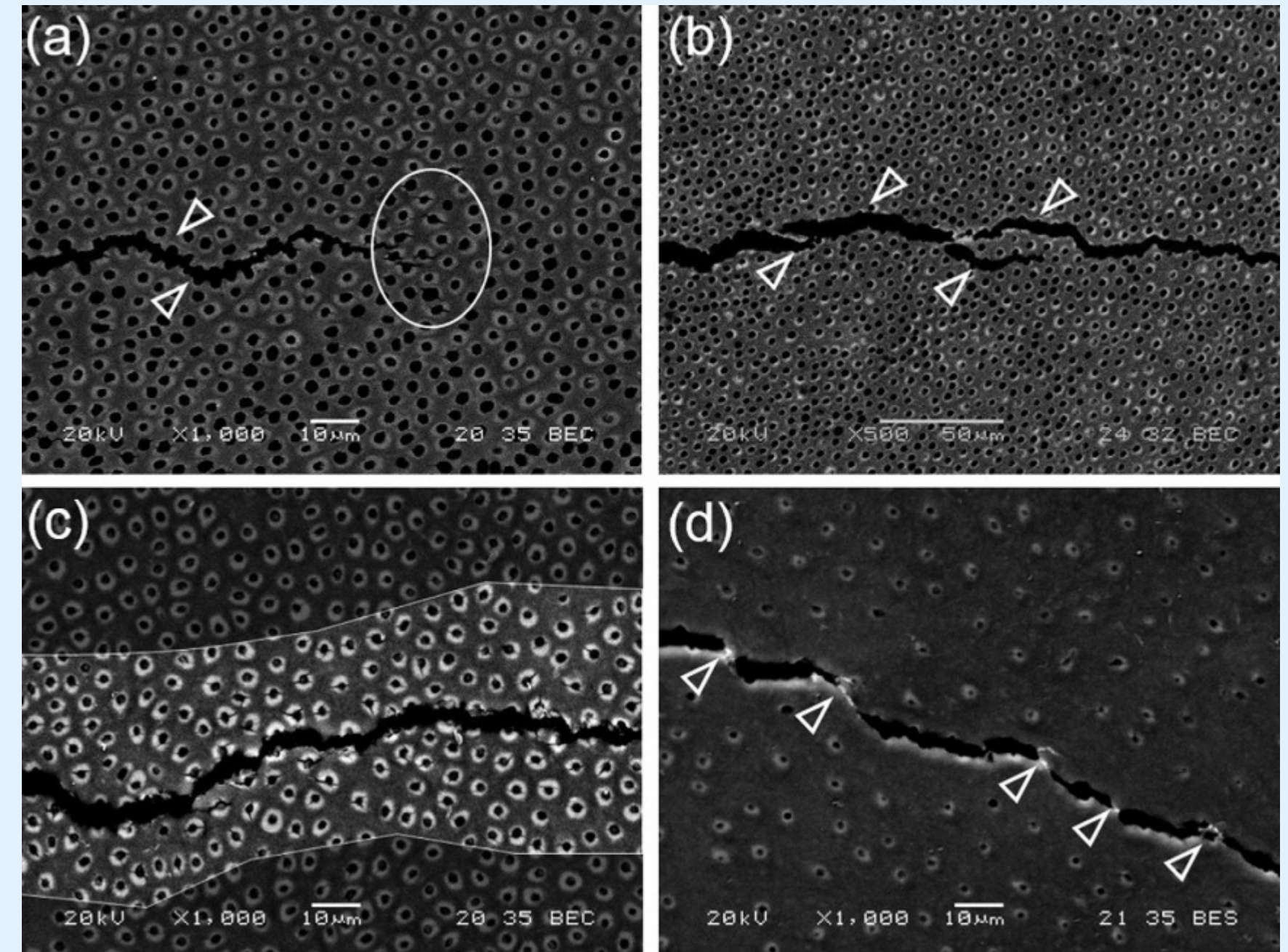
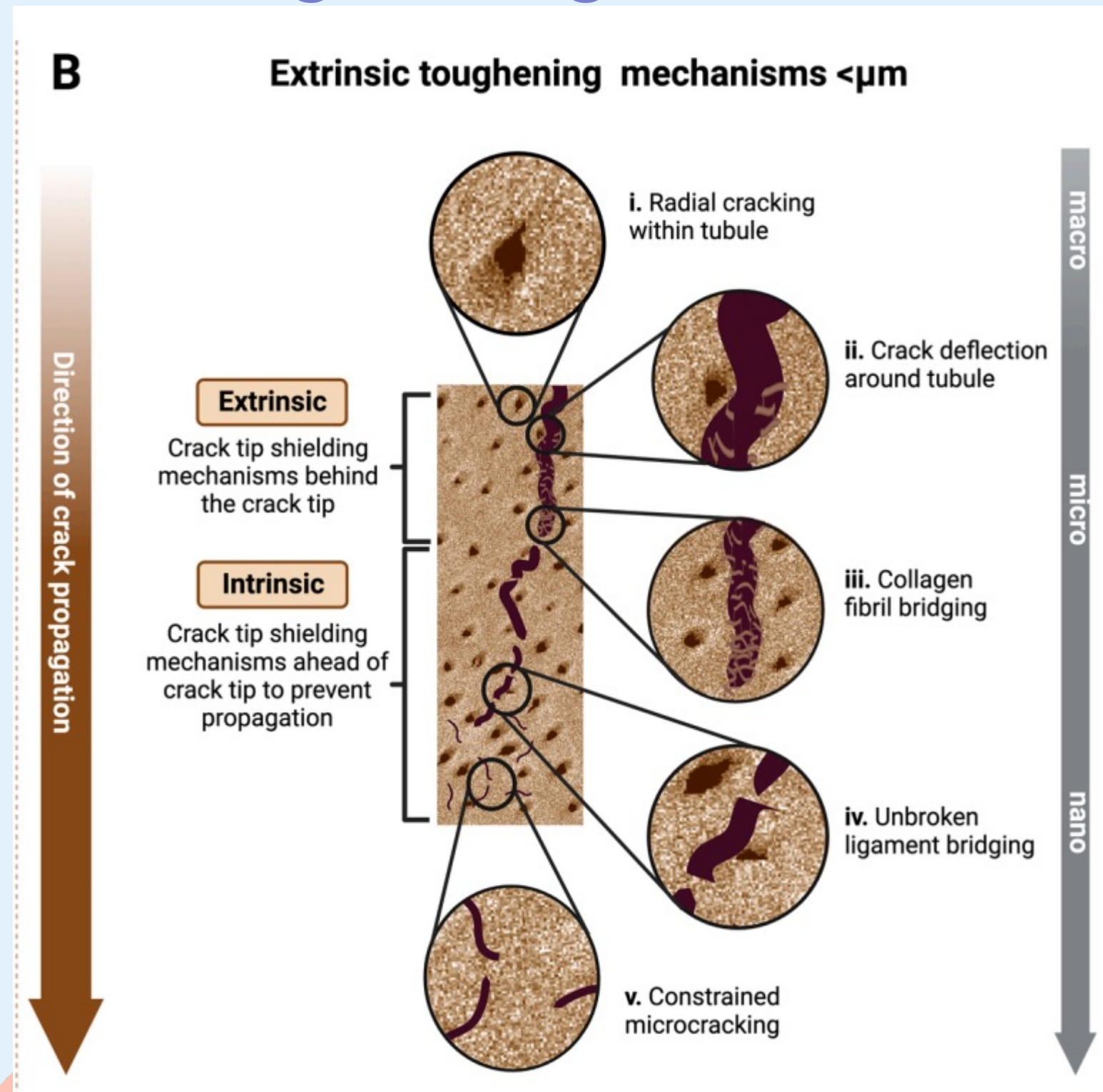
Mechanical Loading

- **Fracture resistance of teeth**
 - Tooth fracture is most likely to originate from cracks that initiate in the dentin
 - Cracks can be introduced to dentin during restorative process or from stress caused by restoration geometry
- **Direction-dependence** fracture resistance
 - More prone to fractures perpendicular to dentin tubules
- **Location-dependence**
 - Increased chance of fracture with increased restoration depth



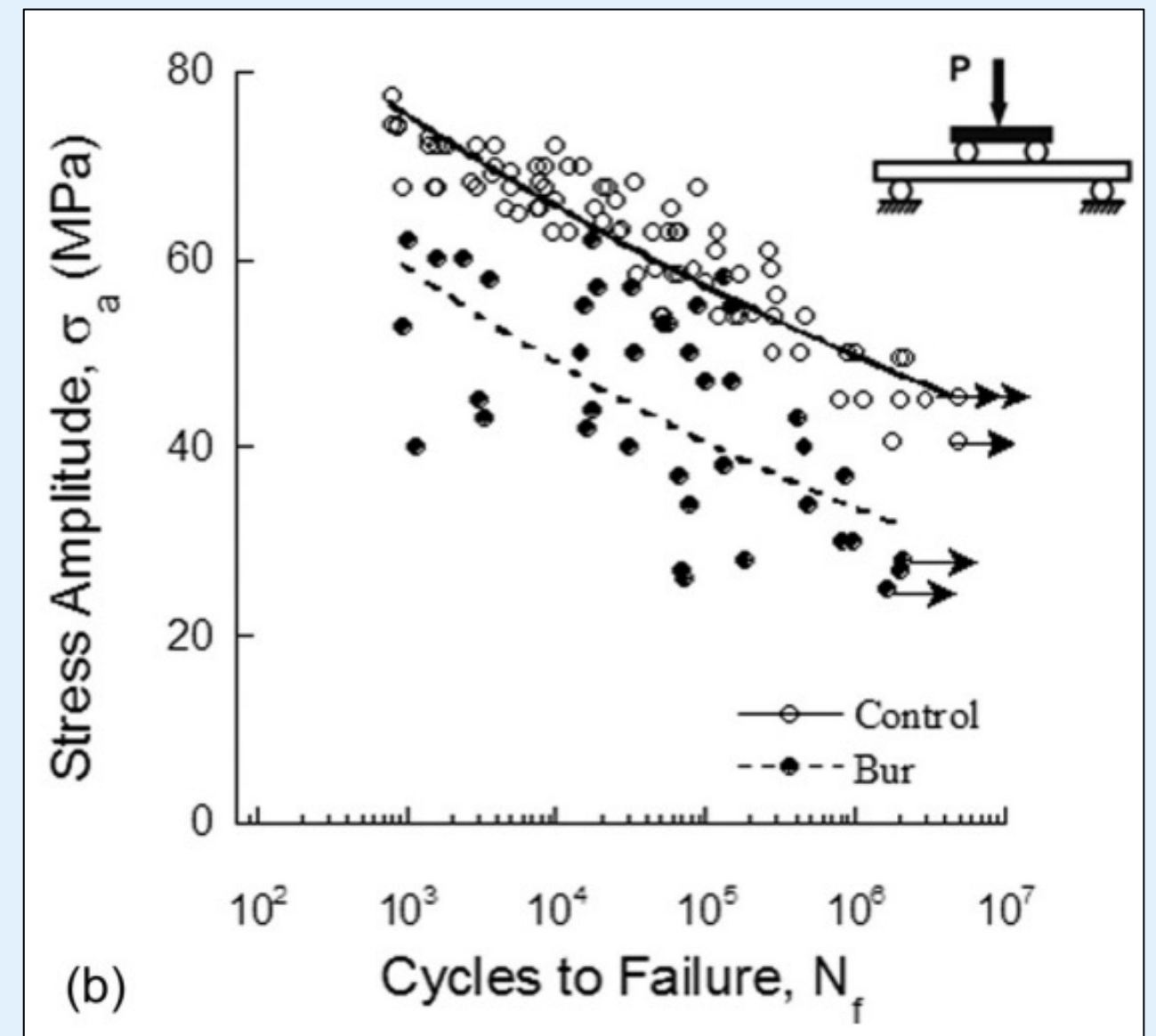
Mechanical Loading

- Toughening mechanisms



Mechanical Loading

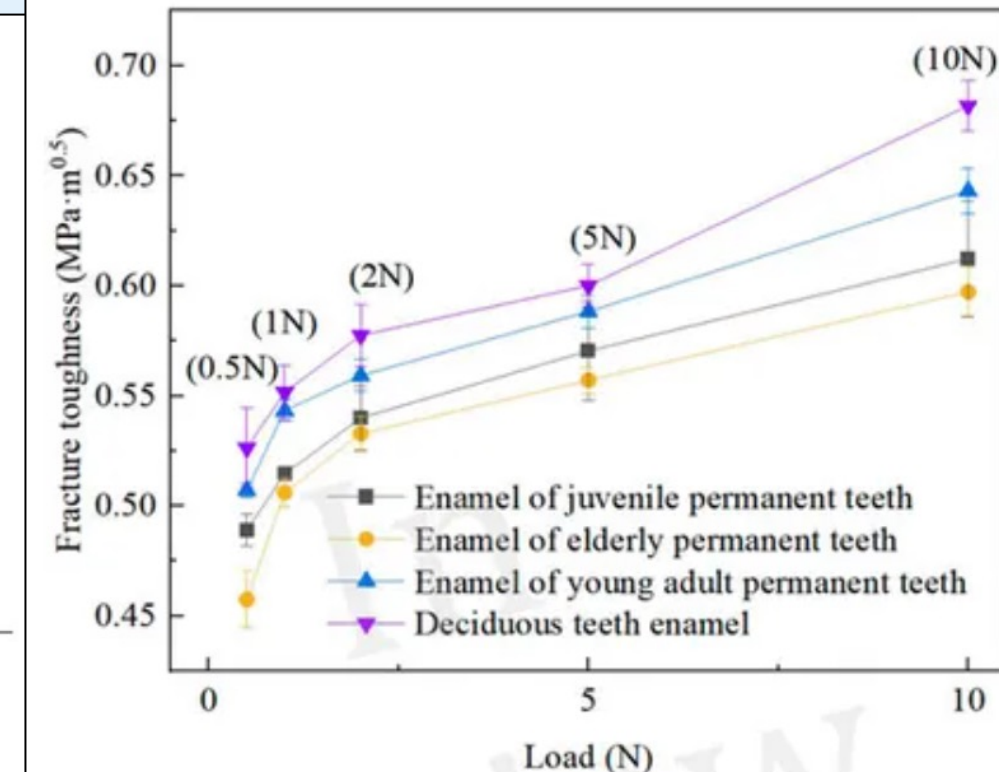
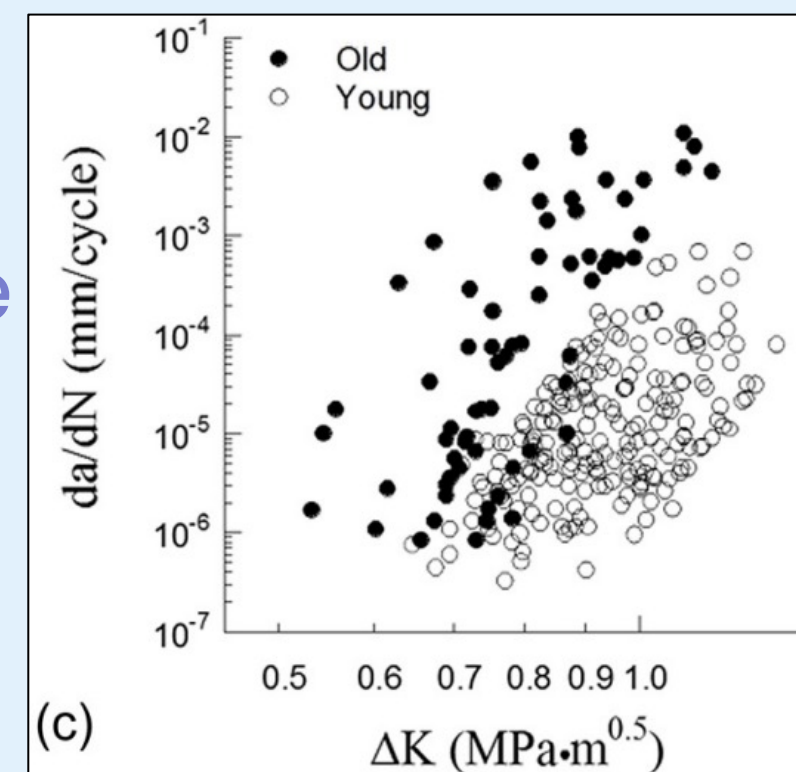
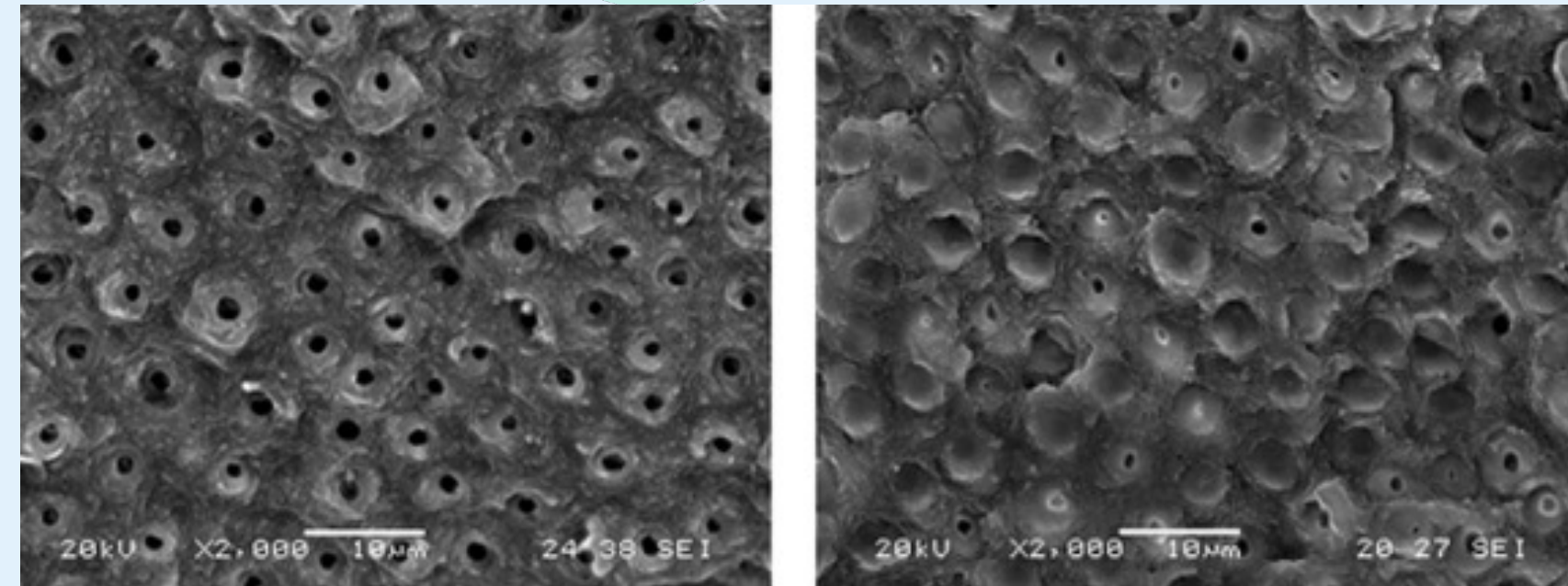
- **Fatigue properties of tooth tissue**
- Bonded resins rods to enamel
 - All fatigue failures occurred in the enamel and adjacent to the adhesive interface
- **Cutting dentin** with carbide burs decreased stress-life fatigue resistance by nearly 40%
- Extrinsic toughening mechanisms are much less effective during **cyclic loading** when compared to the static condition



Aging



- **Dehydration** causes a reduction in the crack growth resistance in dentin
 - low salivary flow with age/medications
- Dentin tubules are **filled with mineral** with age
- Significant reduction in resistance to fatigue crack growth initiation and increase in the rate of incremental extension
- Extrinsic toughening mechanisms decrease with age
- Loss of protein in enamel reduces fracture toughness



Thermal Cycling

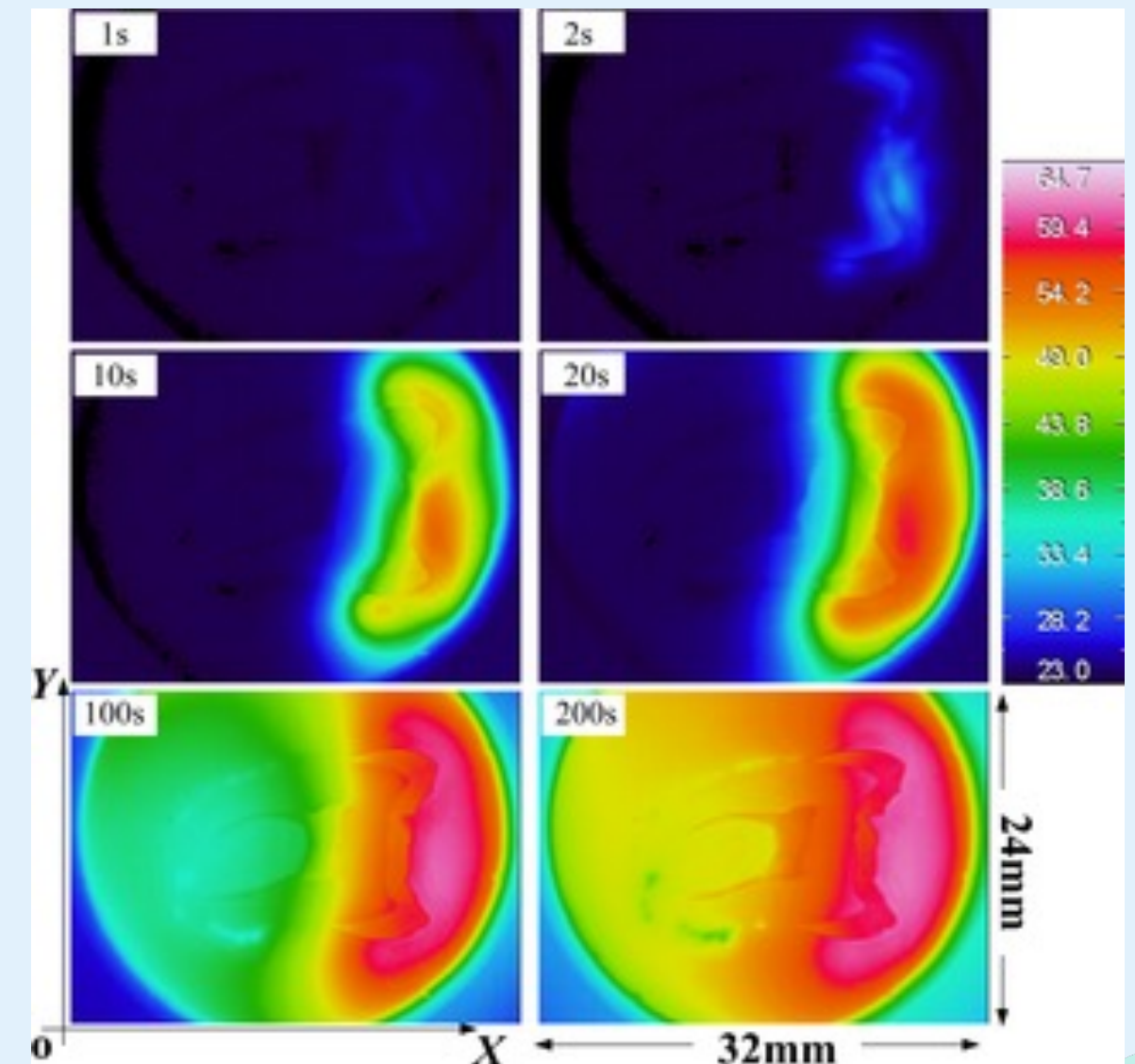
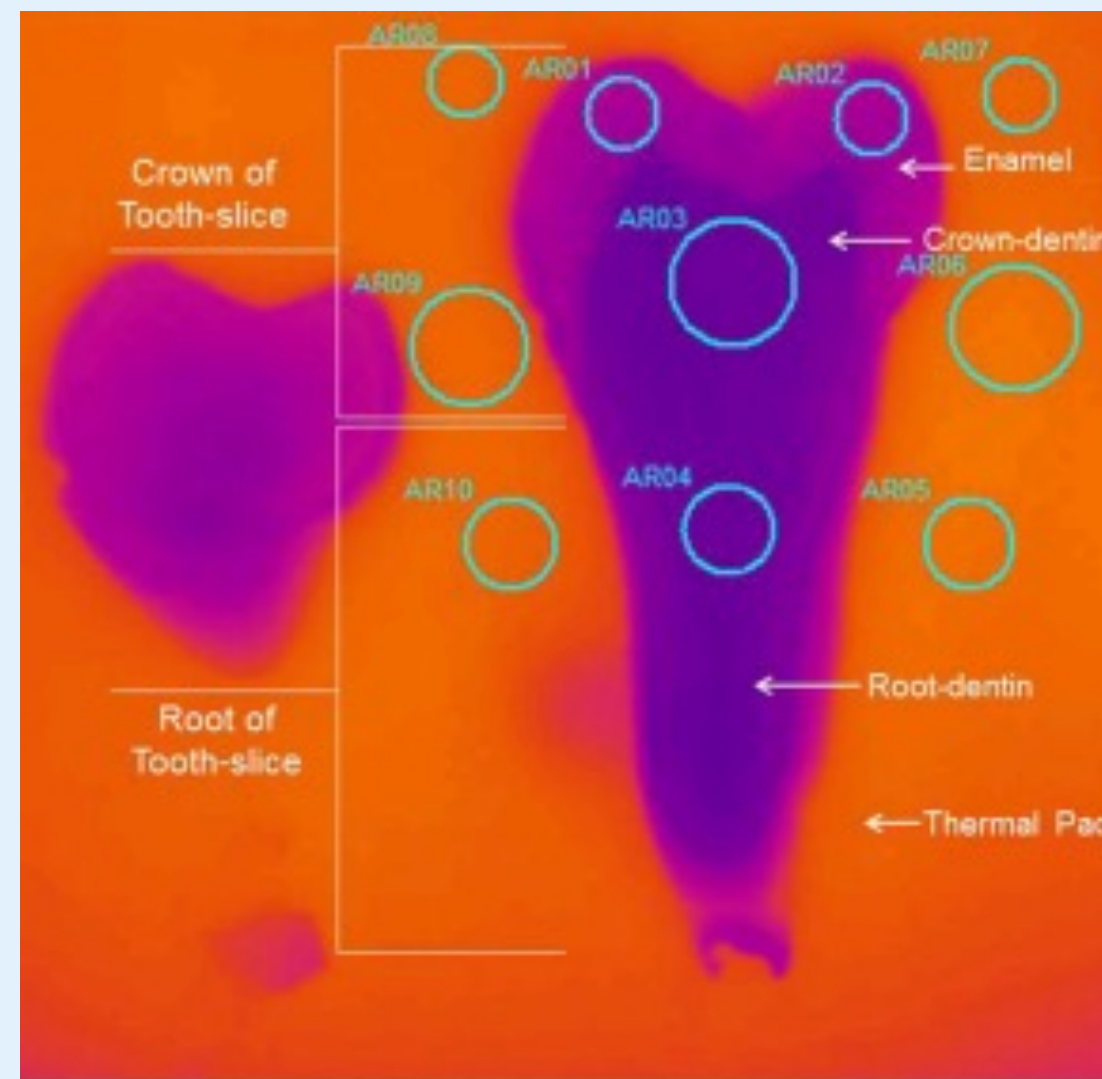
Enamel	Dentin
$16.96 \times 10^{-6}/^{\circ}\text{C}$	$10.59 \times 10^{-6}/^{\circ}\text{C}$

- In vitro process of subjecting extracted tooth to temperature extremes found in the oral cavity
 - Mechanical stress induce crack propagation
 - Changes in gap dimensions
- Collagen degradation
- Affects
 - Fracture toughness
 - Elastic modulus
 - Microhardness



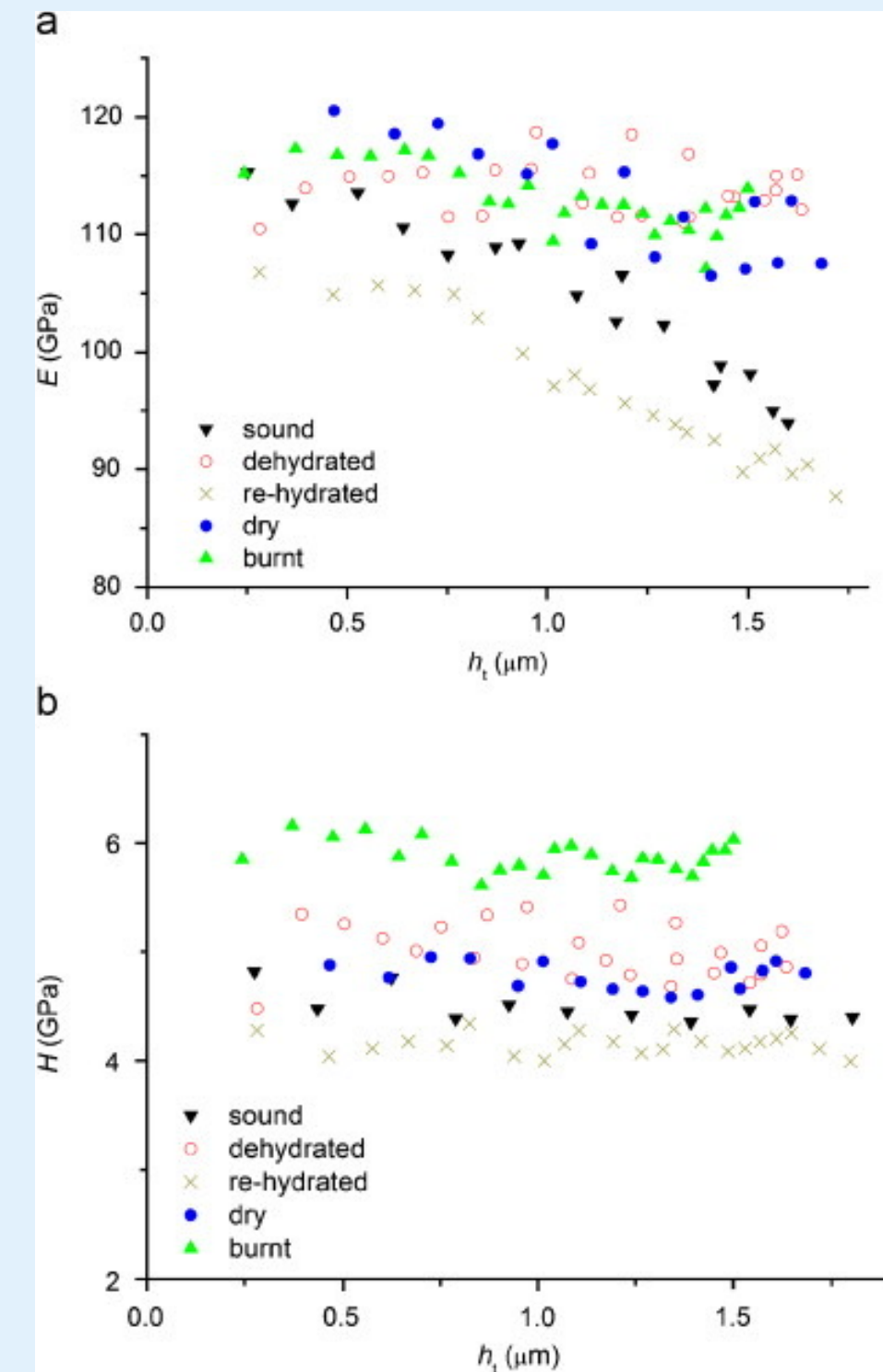
Thermal Cycling

- Thermal buffering by enamel
- Rod structure disperses energy laterally



Xerostomia

- Leads to dehydration of the tooth
- Increased risk for caries and microfractures
- Increases
 - Stiffness
 - Hardness
- Decreases
 - Toughness



Treatments	Shear modulus (GPa)	Hardness (GPa)
Sound	0.114	4.49±0.15
Dehydrated	0.165	5.01±0.25*
Re-hydrated	0.077	4.15±0.10*
Dry	0.164	4.78±0.13*
Burnt	Nil	5.94±0.34*

Requirements for Restorative Materials

Environment	Tooth-Bound Properties	Material Requirements
Acidic Stress	Resists demineralization	Acid resistant/low solubility. Chemically stable Ability to adhere in acidic environments
Mechanical Stress	Enamel: wear resistance, fracture under load Dentin: Cushions enamel to dissipate stress	Enamel: High hardness, fracture toughness, wear resistance Dentin: Flexural strength and lower elastic modulus comparatively
Thermal Stress	Maintains structural integrity. Minimal microfractures	Coefficient of thermal expansion (CTE) Lower thermal conductivity for sensitivity concerns
Xerostomia	Hydration; prevention against demin and caries	Fluoride release or buffer acids. Resistant to crack propagation in low-moisture environments. Maintain bond in dry conditions

BEHAVIORAL OBJECTIVES

4. Compare and contrast restorative materials with natural teeth across these property domains, evaluating where they align, where they diverge, and how these matches or mismatches impact clinical outcomes such as fracture resistance, wear, esthetics, and biocompatibility.

Behavioral Objective 4

Search Strategy

P: Permanent human teeth requiring restoration

I: Restorative materials

C: Natural tooth structure or restorations that mimic natural tooth properties

O: Fracture resistance, wear, esthetics, biocompatibility

Databases

PubMed

Google Scholar

NCBI

Keywords: "fracture" "wear" "biocompatibility" "mechanical properties"

Multiple restorative materials

Results:

- 13,800 unfiltered
- 7,500 filtered for date range 2014–2025
- 300 filtered for reviews

Highest level of evidence: systematic review with meta-analysis

Resin Composite

Mechanical Properties

- Elastic modulus (8–20 GPa) and Hardness (KHN 50–120) similar to dentin, significantly lower than enamel (Chun, K., Choi, H., & Lee, J., 2014)
- Comparable compressive strength to enamel and dentin (250–400 MPa)
- Higher flexural strength than enamel and dentin (100–200 MPa)
- Fracture toughness (0.8–1.6 MPa) is similar to enamel (0.7–1.3) but lower than dentin (2–3)
 - Explains marginal chipping



Resin Composite



- Coefficient of thermal expansion is 25–55 ppm/°C
 - Expands **2–3x** more than natural tooth structure under temperature change
- Composites wear at a rate of 10–50 $\mu\text{m}/\text{year}$, similar to enamel (15–29) and higher than dentin (3–10)
- Translucency of composite varies with material
 - Undergoes color shift over time

Amalgam

- Less stiff (27 GPa) than enamel (70 GPa), stiffer than dentin (18 GPa)
- Similar compressive strength (~297 MPa)
- Stronger than enamel in tension but much weaker than dentin
- Amalgam (110 KHN) is far softer than enamel (350 KHN)
- Similarly brittle ($0.8 \text{ MPa}\sqrt{\text{m}}$) to enamel ($0.7 \text{ MPa}\sqrt{\text{m}}$)



Amalgam

- Unesthetic
 - Limited to posterior restorations
- Similar wear rate to enamel (10–15 $\mu\text{m}/\text{yr}$)



Resin Modified Glass Ionomer

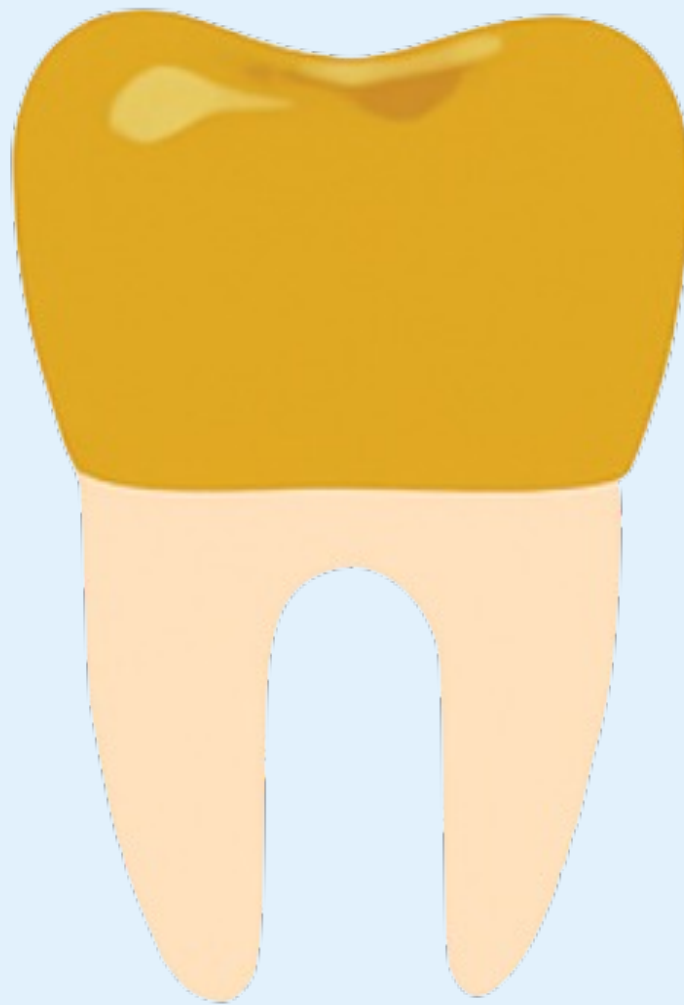


- RMGI is far more compliant (4-6 GPa) → good for cervical lesions
- Significantly weaker (150-220 MPa) than enamel (~384 MPa) and dentin (~297 MPa)
- RMGI $\approx$ dentin hardness; much softer than enamel → higher wear
- RMGI wears **10–15× faster** than enamel
- 5-20 ppm/day Fluoride release
- Very close thermal expansion match
 - → low marginal stress

Direct Materials

Material	Mechanical Properties	Wear & Surface Property	Esthetics	Biocompatibility
Natural Tooth (Chun, 2014)	Enamel: harder and more brittle. Dentin: less stiff and more flexible. Pulp: softest layer.	Enamel: High stiffness, hardness, toughness, viscosity, compressive and shear strength	Chroma, hue, and value are major contributors. Texture and luster Tooth shape and position in the smile	Natural part of the human body Gold standard Great energy dissipation
Composite Resin (Kim, 2022) (Ramos, 2024)	Lower stiffness than enamel High flexural strength	High wear resistance High tensile strength	Highly esthetic, translucency, opalescence and shade matching qualities	Considered biocompatible after polymerization
Amalgam (Qurat-ul-ain, 2021)	High compressive strength Low tensile and shear strength	Microfractures and dislodgement of material	Poor esthetics due to metallic coloring and staining	Allergic rxns rare but possible. Mercury release is minimal.
Resin Modified Glass Ionomer (Ivica,, 2024) (Dursun, 2016)	Increased flexural and tensile strength	Increased hardness and fracture resistance than enamel	Shade matching and translucency	Reduced biocompatibility due to HEMA release

Gold

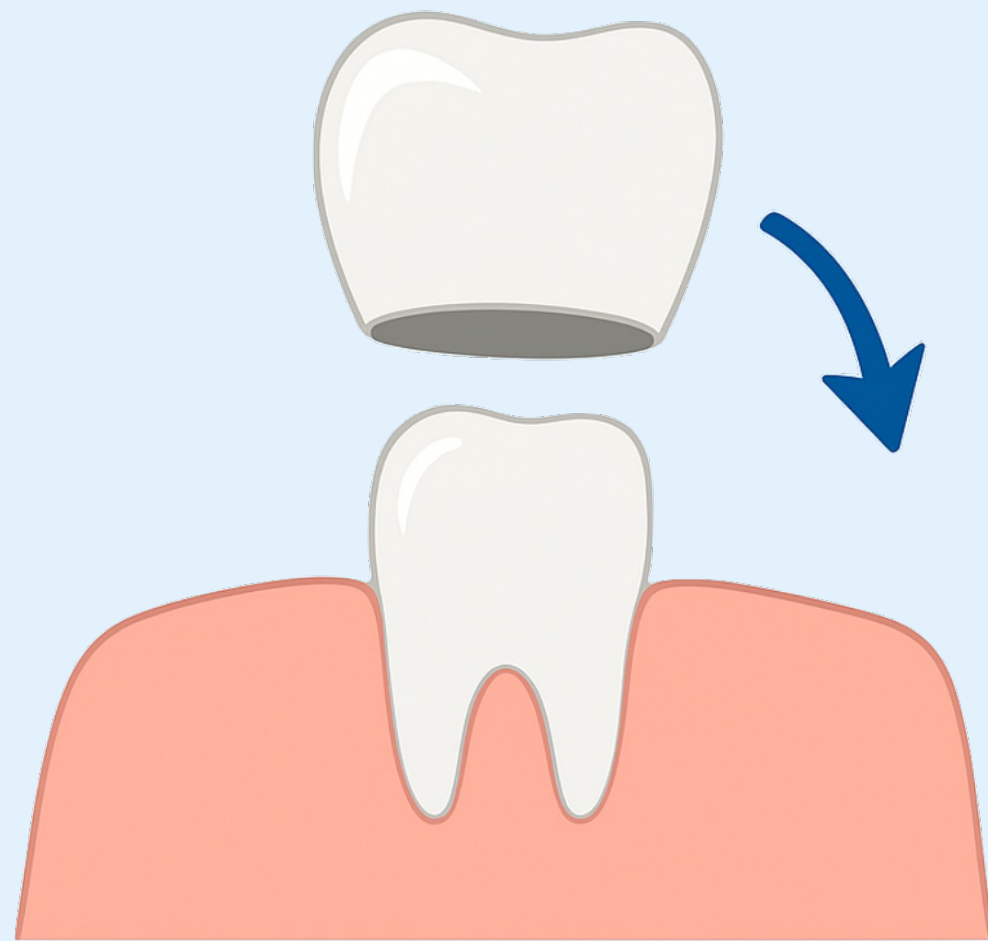


- Gold $\approx$ Enamel in stiffness (90–105 GPa vs 70–110 GPa)
- **500–900 MPa $\rightarrow$ much stronger** than enamel/dentin
- MUCH softer than enamel and softer than ceramics
- Gold wear rate $\approx$ **equal or slightly lower than enamel**
- Gold CTE 14–15 ppm/ $^{\circ}$ C is close to enamel (11–17)

Craig's Restorative Dental Materials, 14th Ed.

Zhang Y, Lawn B. Enamel and dentin mechanics. J Dent Res

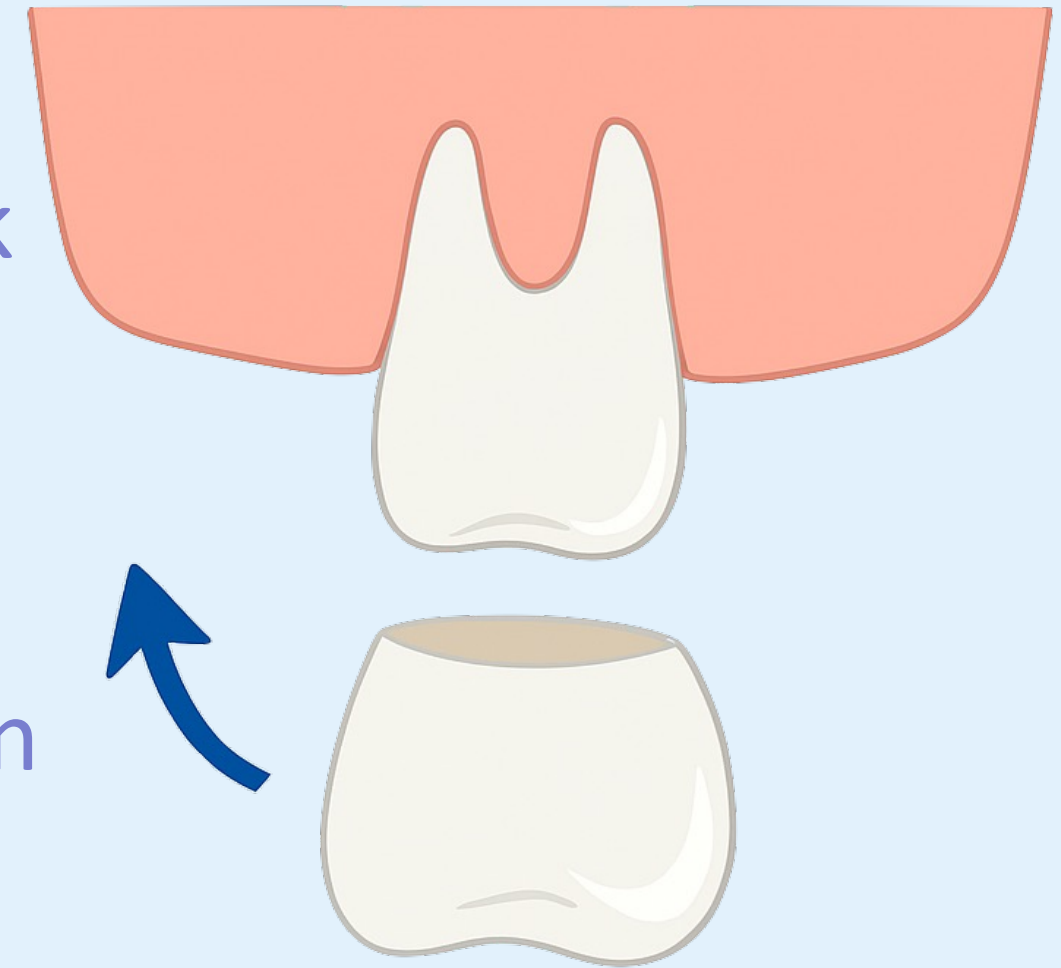
Porcelain Fused to Metal



- PFM crowns behave similarly to an enamel shell on a rigid substrate
- Metal coping fracture toughness ($6\text{--}15 \text{ MPa}\cdot\text{m}^{0.5}$) far exceeds enamel (0.7) and dentin (2–3)
 - Porcelain very low toughness
- Porcelain wears opposing enamel aggressively
- Porcelain $\approx$ translucent but opaque core layer can reduce light transmission

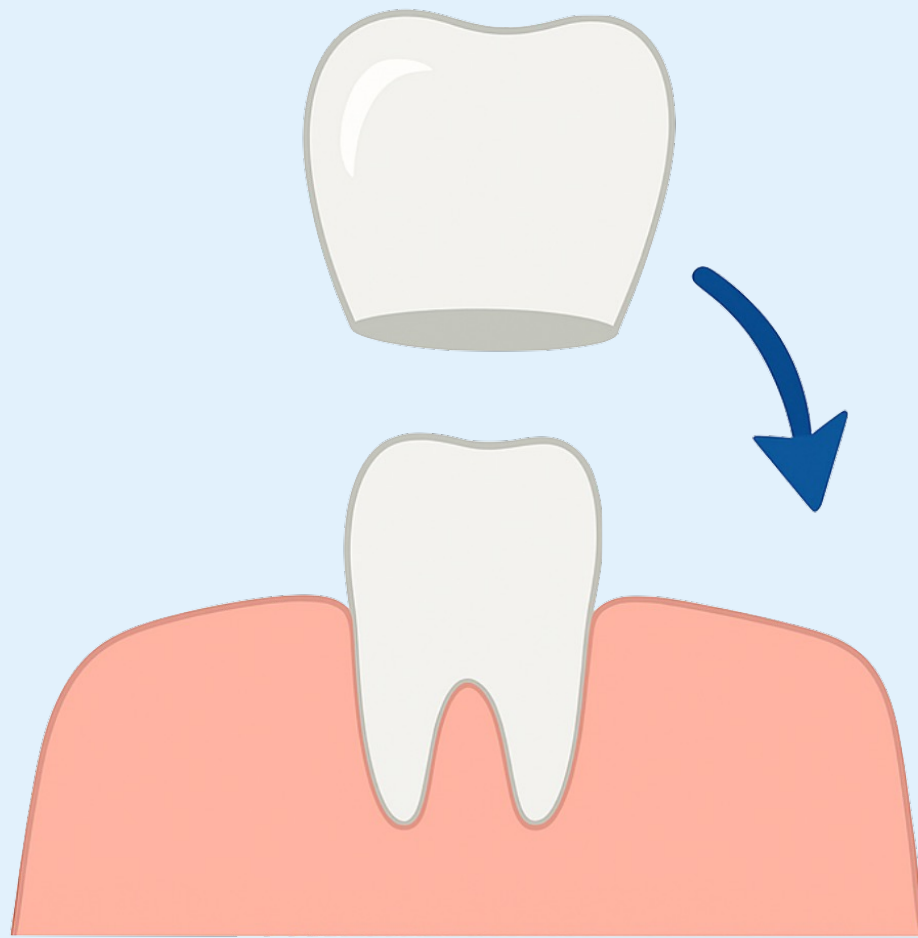
Lithium Disilicate

- Lithium disilicate mimics dentin's resistance to crack propagation
- Very close elastic modulus match → provides good load distribution
- Lithium disilicate is much more enamel-friendly than zirconia
 - Wear rate approaches natural enamel
- More abrasive than natural enamel
- CTE very close to enamel → good margin stability



Elsaka SE. (2014). *Bond strength of novel CAD/CAM lithium disilicate glass-ceramic.*
Journal of Prosthodontics

Zirconia



- Zirconia (210 GPa) is **3× stiffer than enamel** and **10× stiffer than dentin**
- **Compressive (>2000 MPa)** and **flexural strength (~1000 MPa)** massively exceed natural tooth
- **Won't chip easily** like glass ceramics
- **If unpolished**, it can be *more abrasive to enamel* than lithium disilicate or porcelain

Zhang Y, Lawn BR. Novel zirconia materials in dentistry. J Dent Res. 2018;97(2):140–147.

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Indirect Materials

Material	Mechanical Properties	Wear & Surface Property	Esthetics	Biocompatibility
Gold (Chun, 2014)	High tensile strength, hardness, and fracture resistance	Excellent durability and longevity. Minimal wear on opposing teeth	Metallic color does not match the natural shade of teeth.	Highly biocompatible. Corrosion and plaque resistant. Similar thermal expansion coefficient.
Porcelain Fused to Metal (Canadian Agency for Drugs and Technologies in Health, 2015)	High strength, durability, and fracture resistance	Wear on the porcelain surface	Porcelain mimics the enamel Metal collar has poor appeal	Depends on the metal alloy, but most are considered good. Corrosion resistant
Lithium Disilicate (Tafari, 2025)	High flexural strength, fracture toughness, and fatigue resistance	Comparable to enamel but depends on smoothing and polishing of surface	High translucency like enamel. Stain resistant	High chemical stability. Rare allergic reactions.
Zirconia (Canadian Agency for Drugs and Technologies in Health, 2015) (Tafari, 2025)	High strength, fracture toughness, and hardness	Wear depends on surface finish: polishing is crucial	Natural appearance. Color stability.	Stable – nonreacting surrounding tissues. Low affinity for plaque.

BEHAVIORAL OBJECTIVES

5. Apply an evidence-based decision framework to select restorative materials and techniques that best approximate the multidimensional properties of teeth, integrating biological, mechanical, chemical, and patient-specific considerations.

Behavioral Objective 5

Search Strategy

P: Teeth requiring restoration

I: Restorative materials

C: Natural tooth structure

O: Approximation of tooth properties

Databases

PubMed

NCBI

Keywords: "fracture toughness"

"reminerlization" "biomimetics" "biocompatibility"

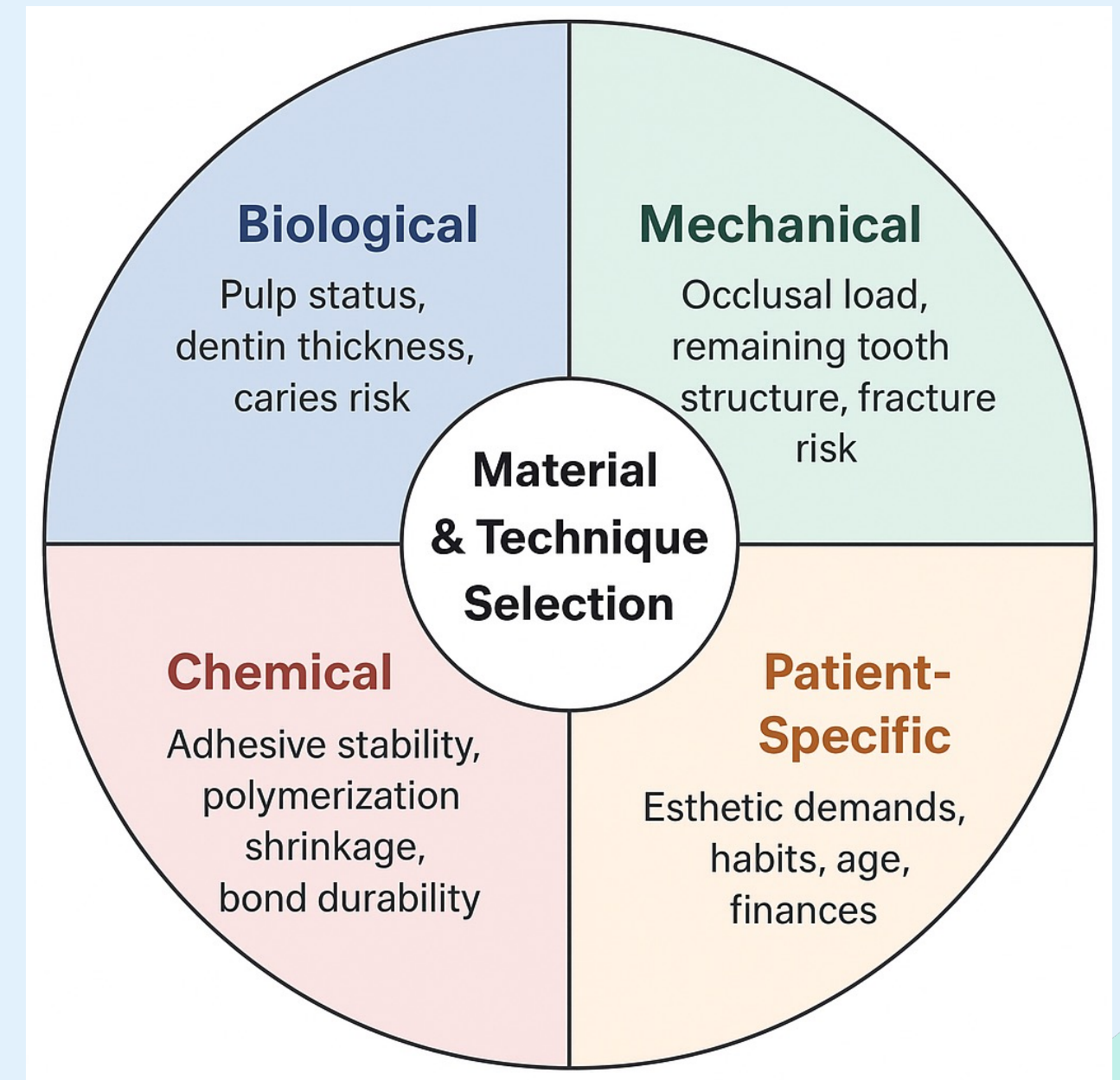
"mechanical properties"

Highest level of evidence: systematic

review with meta-analysis

Evidence-Based Restorative Material Selection

- Integrates mechanical, chemical, biological, and patient-specific factors
- Synthesizes key principles:
 - Tooth structure and properties
 - Restorative materials
 - Patient considerations and clinical outcomes



Mirroring Tooth Properties: Considerations

Mechanical:

- Enamel– high elastic modulus, hard and wear resistant
 - *If too soft* – poor durability and protection of underlying dentin
 - *If too hard* – excessive wear of opposing dentition
- Dentin– flexible, stress absorption, lower elastic modulus
 - *If modulus too low* – can cause stress concentration and marginal breakdown
 - *If modulus too high* – no cushioning and crack propagation
- Maximum stress and strain should also be greater than or equal to enamel/dentin or else restoration may fail internally
- C-factor, thermal expansion and contraction

Mirroring Tooth Properties: Considerations

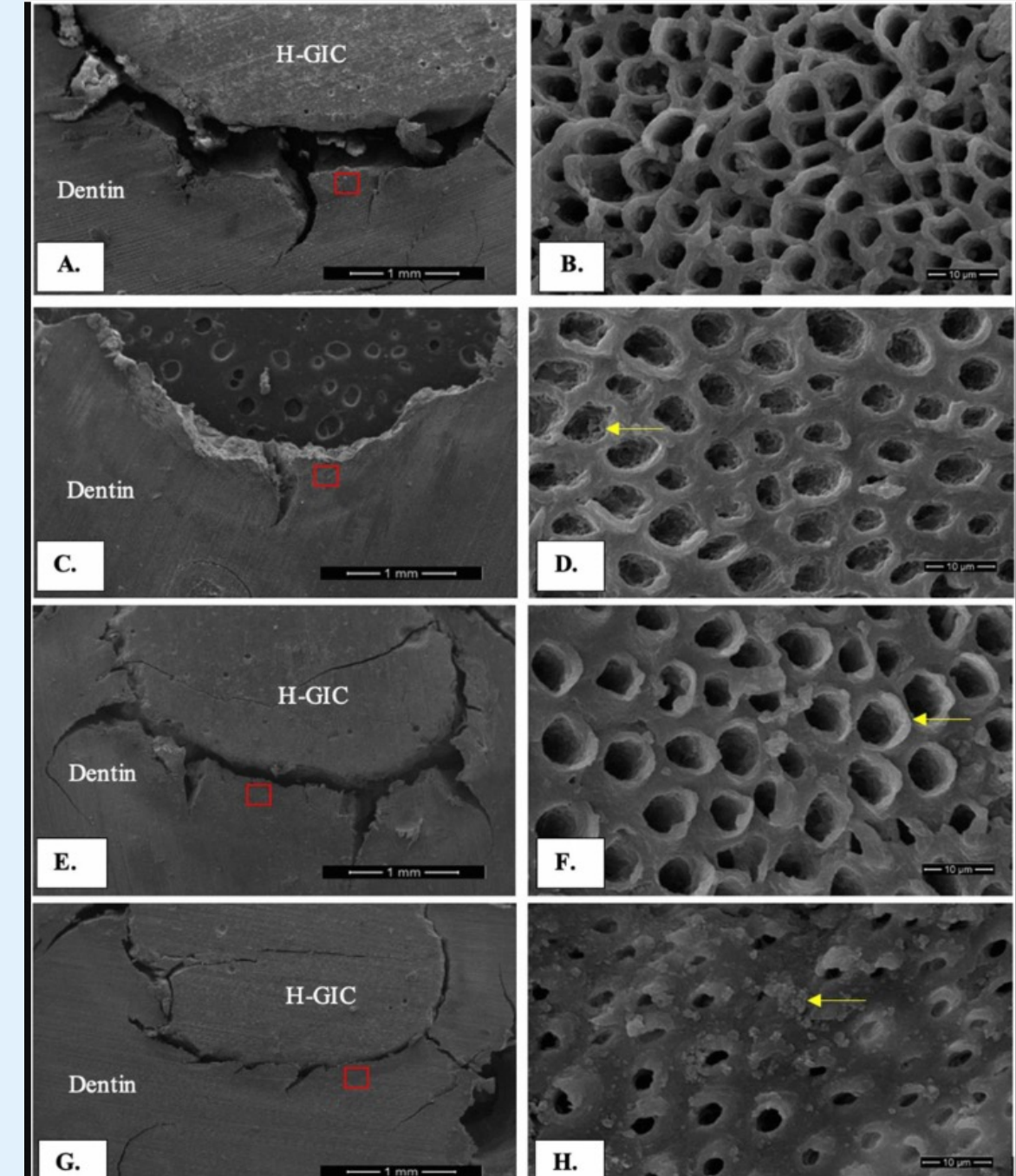
Mechanical (cont.):

- Indirect restorative materials may be required to stabilize teeth and better distribute occlusal forces when:
 - Extensive loss of tooth structure
 - Remaining cusps are thin/undermined
 - Endodontically treated (posterior)
 - Cracks/undermining fracture lines
- Direct restorative materials should be considered when:
 - Tooth structure is relatively intact
 - Low-moderate occlusal stress without parafunctional habits
 - Small fractures on anterior teeth
 - Endodontically treated (anterior)
 - Core build up is required for indirect restoration

Mirroring Tooth Properties: Considerations

Chemical:

- Acid resistance
 - Important for erosive patients (GERD)
 - Sclerotic dentin of eroded teeth exhibit reduced resin bond strength¹
 - GIC and bioactive materials will contribute to local remineralization through the release of various ions²
 - Indirect materials, like gold and zirconia, are highly acid-resistant
- Bonding
 - Chemical bonding of GIC permits moist field, though too dry and strength and esthetics will be compromised³
- Materials should be matched to patient's oral environment



Dentin remineralization following H-GIC restoration contact with or without CHX. **A, B** Equia™ **C, D** CHX-Equia™ **E, F** Ketac™ **G, H** CHX-Ketac™.

Mirroring Tooth Properties: Considerations

Biological:

- Sensitivity related to hydrodynamic theory may result from...
 - Amalgam – high thermal conductivity¹
 - Polymerization shrinkage or poor prep design – void, microleakage¹
- Matrix metalloproteinases (MMPs)
 - Activated by acidic environments and reduce restorative bond strength following their degradation of hybrid layer²
 - MMP inhibitors like those in chlorhexidine combat these effects²
- Caries prevention
 - Ca²⁺-, PO³⁻-, and F-releasing materials for compromised patients³
 - Indirect restorations less likely to have microleakage and bacterial penetration than direct⁵

Mirroring Tooth Properties: Considerations

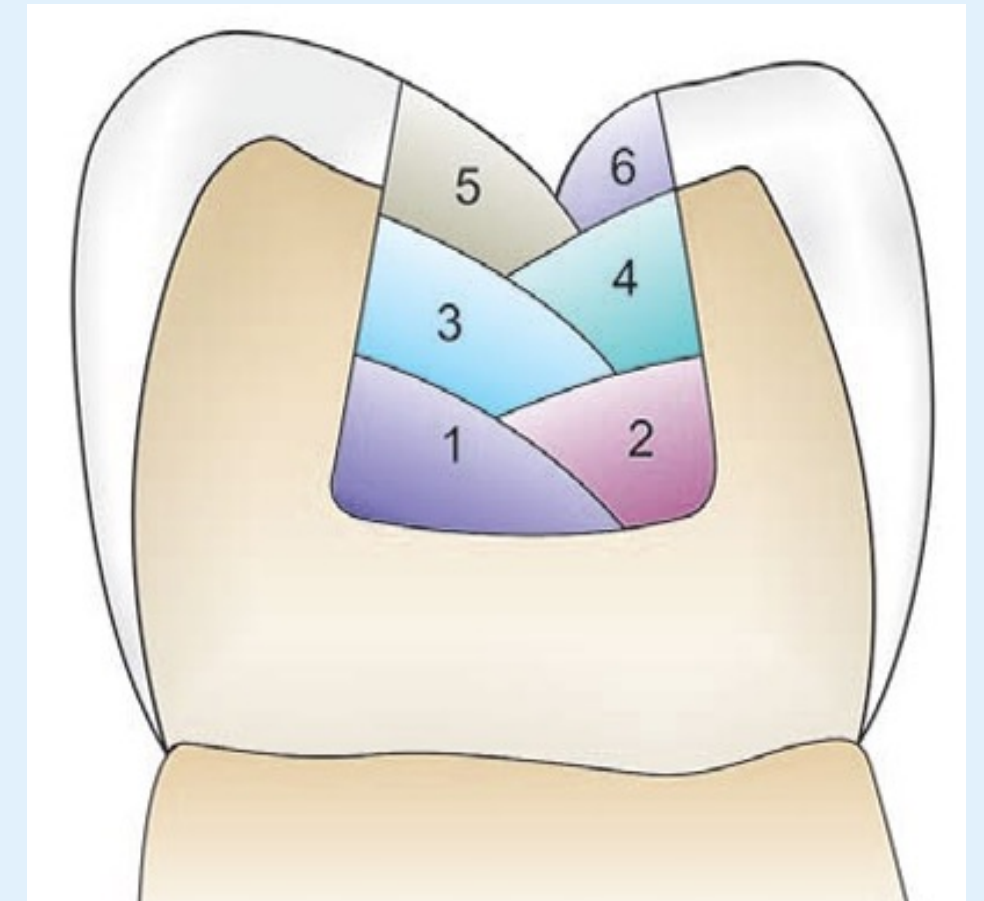
Patient-Specific Factors:

- Age
 - Larger pulp in younger teeth, aim for minimally invasive restoration¹
- Medical history and oral hygiene
 - Prioritize prevention over esthetics
- Esthetic demands
 - Exacting vs. Indifferent personalities
- Finances
- Must tailor restorative approach to each patient



Mirroring Tooth Properties: Techniques

- **Minimizing polymerization shrinkage stress** through incremental layering, "soft start" light curing, low-shrinkage composite resins/bulk fills
- **Minimizing fractures** through proper prep design (no unsupported enamel), structural tools (pins, post), special consideration to occlusion and patient habits, proper polishing
- **Maximize function and adaptation** through the use of different restorative materials, contouring, retracting, and esthetic tools, quality impressions and lab work
- **Maximize longevity** through proper isolation, surface preparation and conditioning, bonding/cementing, burnishing, polishing, patient education



Mirroring Tooth Properties: Techniques

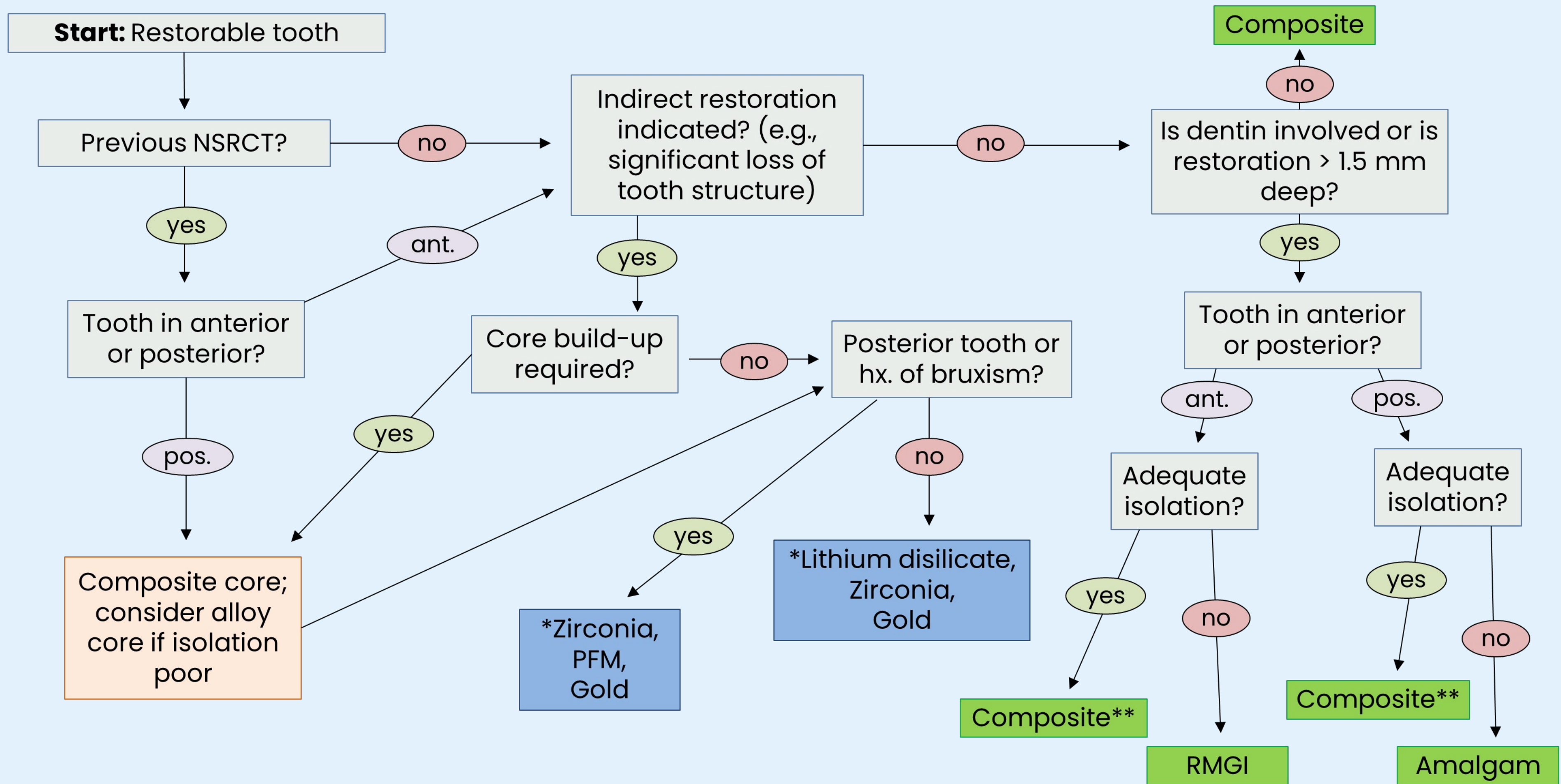
Core Buildup & Crown

- **Objective:** Replicate functional behavior of enamel-dentin complex
- **Core Material:** Replicate dentin
 - Composite → matches elastic modulus, mimics collagen-HA system, reduces fracture risk, bonds reliably to tooth structure
 - Alloy → elastic modulus greater than dentin, reinforce compromised teeth, withstands high functional load, when adhesive bonding is unreliable
- **Crown Material:** Replicate enamel
 - Ceramics → high stiffness, hardness, excellent wear resistance
 - Metals → Ductile, high strength and toughness



Photo: (Nurminsky 2025)

1. **Composite core + Ceramic crown**
2. **Alloy core + Metal/PFM crown**
3. **Composite core + Metal/PFM crown**



*First choice, otherwise patient preference

**Consider RMGI sandwich if sub-gingival

Case Study #1

- 47 year old female presents with CC "My upper right teeth are sensitive to temperature when I eat"
 - Medical Hx: non-contributory
 - Social Hx: 1 drink/week
 - Clinical exam reveals loss of tooth structure apical to the CEJ of teeth #11 & #12
- Which restorative material(s) and techniques would be most appropriate assuming adequate isolation?



FIG. 1

Material: Resin composite with dentin RMGI liner (esthetics + prevention)

- Retention grooves and box form at gingival dentin for retention¹
- Enamel beveling for improved marginal adaptation, increased bonding SA, and esthetics²



❖ Why might it be important to check occlusion?

- Incremental layering and soft-start light curing to reduce polymerization shrinkage³
- Polishing for functional and esthetic purposes⁴

Case Study #2

- 31 year old male presents with CC "I cracked my upper tooth"
- Medical Hx: non-contributory
- Social Hx: denies
- Clinical exam reveals mesio-palatal cusp missing and fracture line below gingival sulcus of #14; PD 3mm
- PA radiograph reveals previous endodontic treatment of #14

➤ Which restorative material(s) and techniques would be most appropriate?



Crown lengthening



Retreatment of root canal



Post and core build up



Removal of old restoration



- BC liner placed after for protection of canals
- Bioactive hybrid resin that releases Ca^{2-} , PO^{3-} , and F
- Good compressive strength and flexural modulus¹

- Dowel space prepared, fiber post placed and tooth built up using dual-cure adhesive and composite (e.g., LuxaCore Z)
- Stress distribution improved through fiber post²
- Compressive strength and flexural modulus of build-up similar to dentin³

Tooth preparation for crown



- Dentin-like feel of build-up material may improve operator precision during preparation

Temporary restoration



- High-strength self-curing bis-acrylic composite (e.g., LuxaCrown) used; good wear resistance and fracture toughness, compressive strength similar to dentin¹
- Occlusion checked and polished after cementation

Delivery of final restoration



- Zirconia chosen for its high strength, fracture toughness and hardness²
- Polished for esthetics and reduction of plaque, fracture risk, and wear on opposing teeth³

BEHAVIORAL OBJECTIVES

- 1. Identify the key properties of natural teeth, including mechanical, physical, chemical, and biological, and explain their role in tooth function and longevity.**

Natural teeth need hardness, elastic modulus, and fracture toughness to properly withstand the functional forces of the dentition. Natural teeth are primarily composed of calcium and phosphate, both protecting against demineralization. Certain trace element substitutions, like fluoride, can decrease solubility of teeth as a protective factor. Natural teeth physical properties include varying shades of translucency and opacity, and their thermal conductivity and expansion values are crucial to consider for restorative material selection. The acquired pellicle serves as a protective mechanism on natural teeth as well.

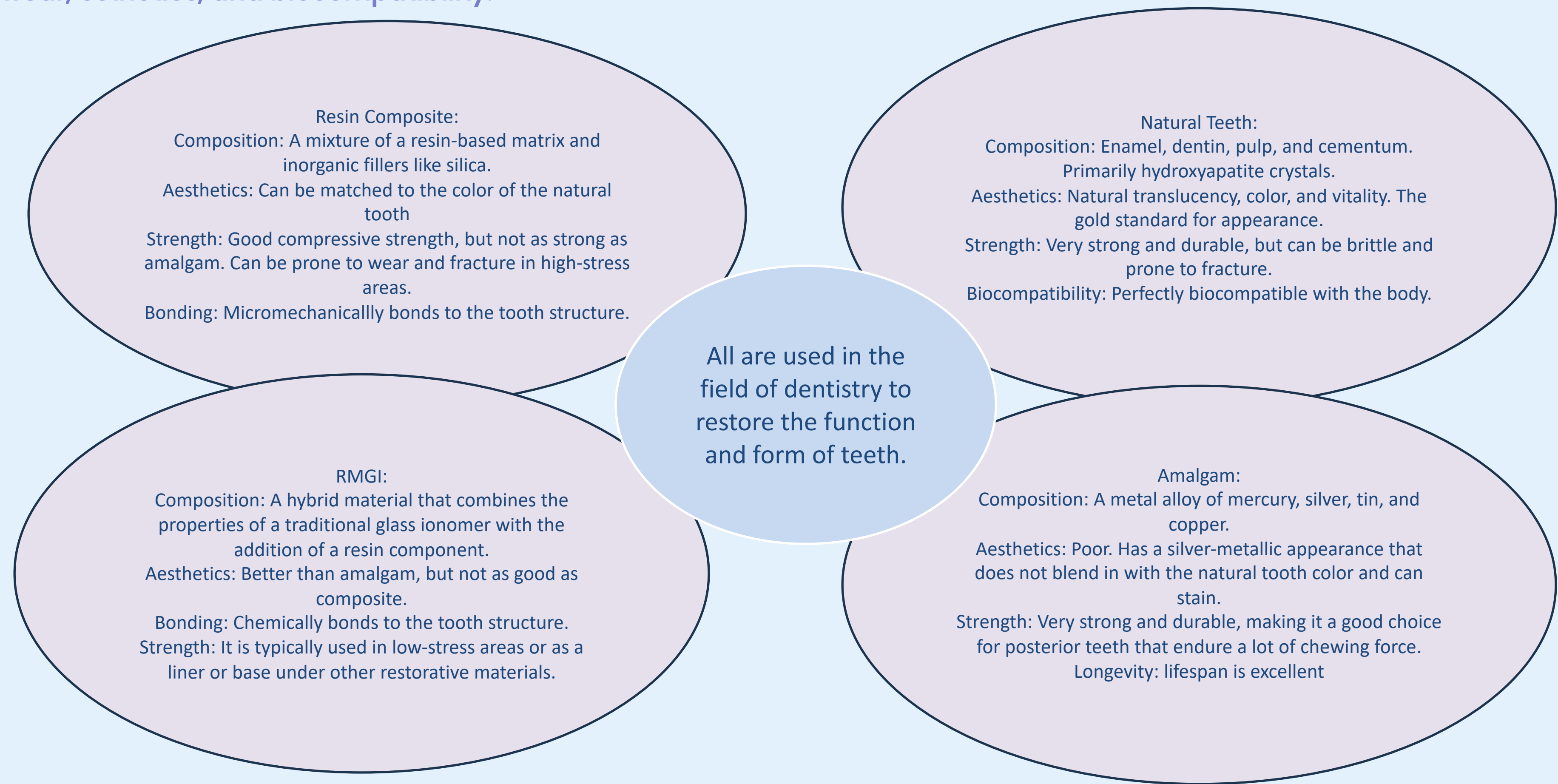
2. Differentiate between enamel and dentin by describing their distinct structural, mechanical, chemical, and biological properties, and explain how these differences influence overall tooth behavior.

Teeth function as a functionally graded composite material: Enamel = stiff, hard shell for wear and protection; Dentin = tough, resilient core that absorbs stress. Their combined properties maximize fracture resistance and longevity under cyclic chewing forces. Different heat coefficients can cause fracture. Dentin is more opaque due to light scattering; enamel more transparent.

3. Explain how natural tooth properties respond to challenges in the oral environment (such as acid attack, mechanical loading, aging, and thermal cycling) and how these responses influence susceptibility to disease and the requirements for restorative material selection.

Dental erosion occurs when acids soften and remove enamel, increasing surface roughness, bacterial adhesion, and overall susceptibility to caries, especially when the pH is under the **critical pH of 5.5 for enamel and 6.8 for dentin**. **Mechanical loading** contributes to wear, cracks, and fatigue, especially in restored teeth with reduced fracture toughness, highlighting the need for materials with high strength, good fracture resistance, and supportive geometry. Enamel naturally deflects cracks and dentin's tubule structure influences resistance, but **aging** reduces these defenses as dehydration, tubule mineralization, and protein loss allow cracks to propagate more easily, increasing fracture risk and requiring materials that can reinforce weakened tooth structure. **Thermal cycling** creates stresses at the DEJ and restoration interfaces due to mismatched thermal expansion, making teeth more prone to leakage and failure, so materials must have compatible coefficients of thermal expansion. **Xerostomia** heightens disease susceptibility by reducing salivary protection, increasing mineral concentration, and compromising adhesion, which demands materials that maintain bond strength in dry, acidic conditions. Together, these challenges weaken tooth structure and elevate risks for caries, fracture, and restoration failure.

4. Compare and contrast restorative materials with natural teeth across these property domains, evaluating where they align, where they diverge, and how these matches or mismatches impact clinical outcomes such as fracture resistance, wear, esthetics, and biocompatibility.



5. Apply an evidence-based decision framework to select restorative materials and techniques that best approximate the multidimensional properties of teeth, integrating biological, mechanical, chemical, and patient-specific considerations.

An evidence-based restorative decision framework starts by selecting materials whose mechanical properties approximate enamel and dentin to provide proper stiffness, stress distribution, and wear resistance without creating rigidity that propagates cracks. Chemical factors—including acid resistance, bonding ability, and ion release—guide choices for patients with erosion or caries risk. Biological considerations such as hypersensitivity, polymerization shrinkage, and MMP activity further narrow material selection. Patient factors like age, esthetic goals, medical history, occlusal habits, and finances determine the final plan and invasiveness. Techniques such as incremental layering, thoughtful prep design, proper bonding, and accurate impressions ensure durable, functional outcomes.

Questions?

Six Thinking Hats®

Kahoot!

[...again](#)



White Hat (Faculty): Provides objective data and scientific evidence about the dental material.



Red Hat (Patient): Expresses emotions, concerns, and preferences regarding the material.



Black Hat (Dentist): Evaluates potential risks, challenges, and clinical implications.



Yellow Hat (Sales Rep): Highlights the benefits and advantages of the material.



Green Hat (Manufacturer): Explores innovation, and future possibilities for the material.



Blue Hat (Board of Dentistry): Ensures all perspectives and standards are considered.

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